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1 Misc

1.1 Template

```
#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace std;

// random
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

// pb_ds stuff
using namespace __gnu_pbds;
const int RANDOM = chrono::high_resolution_clock::now().time_since_epoch().count();
struct chash { int operator()(int x) const { return x ^ RANDOM; }};

template<class T> using ordered_set = tree<T, null_type, less<T>, rb_tree_tag, tree_order_statistics_node_update>;
template<class K, class V> using gmap = gp_hash_table<K, V, chash>;

int main(){
    ios::sync_with_stdio(false);
    cin.tie(0);
}
```

1.2 Random Primes

```
313, 100003, 200003, 1000003, 10000019, 15000017,
2147483647, 1000000087, 1878234273849638459,
2878234273849638443, 8878234273849638443,
91969696969696969, 9206969696969696999
```

1.3 Pragmas

```
#pragma GCC optimize("Ofast")
#pragma GCC optimize("unroll-loops")
#pragma GCC target("sse,sse2,sse3,ssse3,sse4,popcnt,abm,mmx,avx,tune=native")
#pragma GCC target("fpmath=387")
```

1.4 Fast IO

```
int read(){
    char c;
    do {
        c = getchar_unlocked();
    } while(c < 33);
    int res = 0, mul = 1;
    if(c == '-'){
        mul = -1;
        c = getchar_unlocked();
    }
    while('0' <= c && c <= '9'){
        res = res * 10 + c - '0';
    }
}
```

```
c = getchar_unlocked();
}
return res * mul;
}

void write(int x){
    static char wbuf[10];
    if(x < 0){
        putchar_unlocked('-');
        x = -x;
    }
    int idx = 0;
    while(x){
        wbuf[idx++] = x % 10;
        x /= 10;
    }
    if(idx == 0)
        putchar_unlocked('0');
    for(int i = idx-1; i >=0; --i)
        putchar_unlocked(wbuf[i] + '0');
}

void write(constchar* s){
    while(*s){
        putchar_unlocked(*s);
        ++s;
    }
}
```

1.5 Day of Date

```
//0-based
const vector<int> T= {0,3,2,5,0,3,5,1,4,6,2,4};
int day(int d, int m, int y){
    y -= (m < 3);
    return (y + y/4 - y/100 + y/400 + T[m-1] +d) % 7;
}
```

1.6 Number of Days since 1-1-1

```
int rdn(int d, int m, int y){
    if (m < 3) --y, m+=12;
    return 365*y + y/4 - y/100 + y/400 + (153*m - 457)←
        /5 + d - 306;
}
```

1.7 Int to Roman

```
const string R[] = {
    "M", "CM", "D", "CD", "C", "XC", "L",
    "XL", "X", "IX", "V", "IV", "I"
};

const int N[]={
    1000, 900, 500, 400, 100, 90, 50, 40, 10,
    9, 5, 4, 1
};

string to_roman(int x){
    if (x == 0)
        return "0"; //Not decimal 0!
    string res = "";
    for(int i = 0; i < 13; ++i){
        while (x >= N[i])
            x -= N[i];
        res += R[i];
    }
    return res;
}
```

1.8 RNG

```
// RNG - rand_int(min, max), inclusive
mt19937_64 rng(chrono::steady_clock::now().←
    time_since_epoch().count());

template<class T>
T rand_int(T mn, T mx) {
    return uniform_int_distribution<T>(mn, mx)(rng);
}
```

2 Data Structures

2.1 2D Segment Tree

```
struct Segtree2D {
    struct Segtree {
        struct node {
            int l, r, val;
            node* lc, * rc;
            node(int _l, int _r, int _val = INF) : l(_l), r(←
                _r), val(_val),
                lc(NULL), rc(NULL) {}
        };
        typedef node* pnode;
```

```
pnode root;

Segtree(int l, int r) {
    root = new node(l, r);
}

void update(pnode& nw, int x, int val) {
    int l = nw->l, r = nw->r, mid = (l + r) / 2;
    if(l == r)
        nw->val = val;
    else {
        assert(l <= x && x <= r);
        pnode& child = x <= mid ? nw->lc : nw->rc;
        if(!child)
            child = new node(x, x, val);
        else if(child->l <= x && x <= child->r)
            update(child, x, val);
        else {
            do {
                if(x <= mid)
                    r = mid;
                else
                    l = mid + 1;
                mid = (l + r) / 2;
            } while((x <= mid) == (child->l <= mid));
            pnode nxt = new node(l, r);
            if(child->l <= mid)
                nxt->lc = child;
            else
                nxt->rc = child;
            child = nxt;
            update(nxt, x, val);
        }
        nw->val = min(nw->lc ? nw->lc->val : INF,
            nw->rc ? nw->rc->val : INF);
    }

    int query(pnode& nw, int x1, int x2) {
        if(!nw)
            return INF;
        int& l = nw->l, &r = nw->r;
        if(r < x1 || x2 < l)
            return INF;
        if(x1 <= l && r <= x2)
            return nw->val;
        int ret = min(query(nw->lc, x1, x2),
            query(nw->rc, x1, x2));

        return ret;
    }

    void update(int x, int val) {
        assert(root->l <= x && x <= root->r);
        update(root, x, val);
    }

    int query(int l, int r) {
        return query(root, l, r);
    }
};

struct node {
    int l, r;
    Segtree y;
    node* lc, * rc;
    node(int _l, int _r) : l(_l), r(_r), y(0, MAX),
        lc(NULL), rc(NULL) {}
};
typedef node* pnode;
```

```
pnode root;

Segtree2D(int l, int r) {
    root = new node(l, r);
}

void update(pnode& nw, int x, int y, int val) {
    int& l = nw->l, &r = nw->r, mid = (l + r) / 2;
    if(l == r)
        nw->y.update(y, val);
    else {
        if(x <= mid) {
            if(!nw->lc)
                nw->lc = new node(l, mid);
            update(nw->lc, x, y, val);
        } else {
            if(!nw->rc)
                nw->rc = new node(mid + 1, r);
            update(nw->rc, x, y, val);
        }
        val = min(nw->lc ? nw->lc->y.query(y, y) : INF,
            nw->rc ? nw->rc->y.query(y, y) : INF);
        nw->y.update(y, val);
    }

    int query(pnode& nw, int x1, int x2, int y1, int y2) ←
        {
            if(!nw)
                return INF;
            int& l = nw->l, &r = nw->r;
            if(r < x1 || x2 < l)
                return INF;
            if(x1 <= l && r <= x2)
                return nw->y.query(y1, y2);
            int ret = min(query(nw->lc, x1, x2, y1, y2),
                query(nw->rc, x1, x2, y1, y2));

            return ret;
        }

    void update(int x, int y, int val) {
        assert(root->l <= x && x <= root->r);
        update(root, x, y, val);
    }

    int query(int x1, int x2, int y1, int y2) {
        return query(root, x1, x2, y1, y2);
    }
};
```

2.2 Fenwick RU-RQ

```
void updtRL(int l, int r, ll val) {
    updt(BIT1, l, val), updt(BIT1, r + 1, -val);
    updt(BIT2, l, val * (l - 1)), updt(BIT2, r + 1, -val ←
        * r);
}

ll query(int k) {
    return que(BIT1, k) * k - que(BIT2, k);
}
```

2.3 Persistent Segment Tree

```
class PersistentSegtree{
private:
    int n, ptr, sz;
    struct P {
        int val = 0, l, r;
    };
```

```
vector<P> node;
vector<int> root;

int newNode(){
    node.emplace_back();
    return ptr++;
}

int copyNode(int idx){
    node.push_back(node[idx]);
    return ptr++;
}

int build(int l, int r){
    int idx = newNode();
    if (l == r)
        return idx;
    node[idx].l = build(l, (l+r)/2);
    node[idx].r = build((l+r)/2 + 1, r);
    return idx;
}

int update(int idx, int l, int r, int x, int val){
    idx = copyNode(idx);
    if (l == r){
        node[idx].val += val;
        return idx;
    }
    int mid = (l+r)/2;
    if (x <= mid)
        node[idx].l = update(node[idx].l, l, mid, x, val)↵
    ;
    else
        node[idx].r = update(node[idx].r, mid+1, r, x, ↵
        val);
    node[idx].val = node[node[idx].l].val + node[node↵
    [idx].r].val;
    return idx;
}

int query(int idxl, int idxr, int l, int r, int x, ↵
    int y){
    if (y < l || r < x)
        return 0;
    if (x <= l && r <= y)
        return node[idxr].val - node[idxl].val;
    int mid = (l+r)/2;
    return query(node[idxl].l, node[idxr].l, l, mid, x,↵
    y)
    + query(node[idxl].r, node[idxr].r, mid+1, r, x, ↵
    y);
}

public:
PersistentSegtree(int _n) : n(_n), ptr(0){
    sz = 30*n;
    node.reserve(sz);
    root.push_back(build(1,n));
}

void update(int x, int val){
    root.push_back(update(root.back(), 1, n, x, val));
}

int query(int l, int r, int x, int y){
    return query(root[l-1], root[r], 1, n, x, y);
}
};
```

2.4 Segment Tree Beats

```
ll ST1[4*N + 5], ST2[4*N + 5]; //MAX value, SUM Value
ll smax[4*N + 5], occ[4*N + 5]; //2nd Max, occurence of ↵
1st Max
ll arr[N + 5];
```

```
void pushup(int idx){
    ST1[idx] = max(ST1[2*idx], ST1[2*idx + 1]);
    ST2[idx] = ST2[2*idx] + ST2[2*idx + 1];
    if (ST1[2*idx] == ST1[2*idx + 1]){
        smax[idx] = max(smax[2*idx], smax[2*idx + 1]);
        occ[idx] = occ[2*idx] + occ[2*idx + 1];
    }
    else{
        smax[idx] = min(ST1[2*idx], ST1[2*idx + 1]);
        smax[idx] = max(smax[idx], max(smax[2*idx],smax[2*↵
        idx + 1]));
        if (ST1[2*idx] <= ST1[2*idx + 1])
            occ[idx] = occ[2*idx + 1];
        if (ST1[2*idx] > ST1[2*idx + 1])
            occ[idx] = occ[2*idx];
    }
}

void build(int idx, int l, int r){
    if (l == r){
        ST1[idx] = ST2[idx] = arr[l];
        occ[idx] = 1;
        smax[idx] = -1;
        return;
    }
    int mid = (l+r)>>1;
    build(2*idx, l, mid);
    build(2*idx + 1, mid + 1, r);
    pushup(idx);
}

void pushdown(int idx, ll v){
    if (ST1[idx] <= v) return;
    assert(v > smax[idx]);
    ST2[idx] -= (ll)(ST1[idx] - v) * (ll)occ[idx];
    ST1[idx] = v;
}

void down(int idx, ll v){
    pushdown(2*idx, v);
    pushdown(2*idx + 1, v);
}

void update(int idx, int l, int r, int le, int ri, ll v↵
    ){
    //update min(Ai,t)
    if (r < le || ri < l || ST1[idx] <= v)
        return;
    if (le <= l && r <= ri && smax[idx] < v){
        pushdown(idx, v);
        return;
    }
    down(idx, ST1[idx]);
    int mid = (l+r)>>1;
    update(2*idx, l, mid, le, ri, v);
    update(2*idx + 1, mid + 1, r, le, ri, v);
    pushup(idx);
}

ll query1(int idx, int l, int r, int le, int ri){
    //sumquery
    if (r < le || ri < l)
        return -1;
    if (le <= l && r <= ri)
        return ST1[idx];
    down(idx, ST1[idx]);
    int mid = (l+r)>>1;
    ll resa = query1(2*idx, l, mid, le, ri);
    ll resb = query1(2*idx + 1, mid + 1, r, le, ri);
    return max(resa, resb);
}

ll query2(int idx, int l, int r, int le, int ri){
    //maxquery
    if (r < le || ri < l)
        return 0;
    if (le <= l && r <= ri)
```

```
    return ST2[idx];
    down(idx, ST1[idx]);
    int mid = (l+r)>>1;
    ll resa = query2(2*idx, l, mid, le, ri);
    ll resb = query2(2*idx + 1, mid + 1, r, le, ri);
    return resa + resb;
}
```

2.5 Li-Chao Tree

```
// max li-chao tree
// works for the range [0, MAX - 1]
// if min li-chao tree:
// replace every call to max() with min() and every > ↵
with <
// also replace -INF with INF

struct Func {
    ll m, c;
    ll operator()(ll x) {
        return x * m + c;
    }
};

const int MAX = 1e9 + 1;
const ll INF = 1e18;
const Func NIL = {0, -INF};

struct Node {
    Func f;
    Node* lc;
    Node* rc;

    Node() : f(NIL), lc(nullptr), rc(nullptr) {}
    Node(const Node& n) : f(n.f), lc(nullptr), rc(nullptr)↵
    {}
};

Node* root = new Node;

void insert(Func f, Node* cur = root, int l = 0, int r ↵
    = MAX - 1) {
    int m = l + (r - l) / 2;
    bool left = f(l) > cur->f(l);
    bool mid = f(m) > cur->f(m);
    if (mid)
        swap(f, cur->f);
    if (l != r) {
        if (left != mid) {
            if (!cur->lc)
                cur->lc = new Node(*cur);
            insert(f, cur->lc, l, m);
        } else {
            if (!cur->rc)
                cur->rc = new Node(*cur);
            insert(f, cur->rc, m + 1, r);
        }
    }
}

ll query(ll x, Node* cur = root, int l = 0, int r = MAX↵
    - 1) {
    if (!cur)
        return -INF;
    if (l == r)
        return cur->f(x);
    int m = l + (r - l) / 2;
    if (x <= m)
        return max(cur->f(x), query(x, cur->lc, l, m));
    else
```

```
return max(cur->f(x), query(x, cur->rc, m + 1, r));
}
```

2.6 Heavy-Light Decomposition

```
// graph 1-based index
class HLD{
private:
vector<vector<int>> adj;
vector<int> sz, in, out, nxt, par;
Segtree DS;
int timer = -1;

void dfs (int u, int p){
sz[u] = 1;
par[u] = p;
for (auto &v : adj[u])
if (v != p){
dfs(v, u);
sz[u] += sz[v];
if (sz[v] >= sz[adj[u][0]])
swap(v, adj[u][0]);
}
}

void chain(int u, int p){
if(p == u)
nxt[u] = u;
in[u] = ++timer;
for (auto &v : adj[u])
if(v != p){
nxt[v] = (v == adj[u][0] ? nxt[u] : v);
chain(v, u);
}
out[u] = timer;
}

public:
HLD(int n){
adj.assign(n+1, vector<int>());
sz.assign(n+1, 0);
in.resize(n+1);
out.resize(n+1);
nxt.assign(n+1, 0);
par.assign(n+1, -1);
}

void add_edge(int u, int v){
adj[u].push_back(v);
adj[v].push_back(u);
}

void update_path(int u, int v, int x){
for(; in[nxt[u]] != in[nxt[v]]; v = par[nxt[v]]){
if(in[u] > in[v]) swap(u, v);
DS.update(in[nxt[v]], in[v], x);
}
if(in[u] > in[v]) swap(u, v);
DS.update(in[u], in[v], x);
}

int query_path(int u, int v){
int ans = -1;
for(; in[nxt[u]] != in[nxt[v]]; v = par[nxt[v]]){
if(in[u] > in[v]) swap(u, v);
ans = max(ans, DS.query(in[nxt[v]], in[v]));
}
if (in[u] > in[v]) swap(u, v);
ans = max(ans, DS.query(in[u], in[v]));
return ans;
}

void update_subtree(int u, int x){
DS.update(in[u], out[u], x);
}
```

```
int query_subtree(int u){
return DS.query(in[u], out[u]);
}

void init(){
dfs(1, 1);
chain(1, 1);
DS.init(timer + 1);
}
}
```

2.7 Treap

```
// Complexity: O(log N) for split and merge
//
// empty treap: Treap* tr = nullptr;
// insert v at x: [l, r] = split(tr, x), m = Treap(v), ←
// merge lmr
// delete at x: [l, r] = split(tr, x), [m, r] = split(r←
// , 1), merge lr
// lazy prop: propagate every time a node is accessed

mt19937_64 rng(chrono::steady_clock::now().←
time_since_epoch().count());

using Key = int;

struct Treap {
Key val;
Treap* left;
Treap* right;
int prio, sz;
Treap() {}
Treap(int _val);
};

int size(Treap* tr) {
return tr ? tr->sz : 0;
}

void update(Treap* tr) {
tr->sz = 1 + size(tr->left) + size(tr->right);
}

Treap::Treap(Key _val) :
val(_val), left(nullptr), right(nullptr), prio(rng())←
{
update(this);
}

pair<Treap*, Treap*> split(Treap* tr, int sz) {
if(!tr) return {nullptr, nullptr};
int left_sz = size(tr->left);
if(sz <= left_sz) {
auto [left, mid] = split(tr->left, sz);
tr->left = mid;
update(tr);
return {left, tr};
} else {
auto [mid, right] = split(tr->right, sz - left_sz -←
1);
tr->right = mid;
update(tr);
return {tr, right};
}
}

Treap* merge(Treap* l, Treap* r) {
if(!l)
return r;
if(!r)
return l;
```

```
return l;
if(l->prio < r->prio) {
l->right = merge(l->right, r);
update(l);
return l;
} else {
r->left = merge(l, r->left);
update(r);
return r;
}
}
```

2.8 Unordered Map Custom Hash

```
struct custom_hash {
static uint64_t splitmix64(uint64_t x) {
x += 0x9e3779b97f4a7c15;
x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
return x ^ (x >> 31);
}

size_t operator()(uint64_t x) const {
static const uint64_t FIXED_RANDOM =
chrono::steady_clock::now().time_since_epoch().←
count();
return splitmix64(x + FIXED_RANDOM);
}
};

unordered_map<int, int, custom_hash> umap;
```

2.9 Interval Container

```
/'KACTLsIntervalContainer
set<pii>::iterator addInterval(set<pii> &is, int L, int←
R){
if (L == R) return is.end();
auto it = is.lower_bound({L,R}), before = it;
while (it != is.end() && it->first <= R){
R = max(R, it->second);
before = it = is.erase(it);
}
if (it != is.begin() && (--it)->second >= L){
L = min(L, it->first);
R = max(R, it->second);
is.erase(it);
}
return is.insert(before, {L, R});
}

void removeInterval(set<pii> &is, int L, int R){
if(L == R) return;
auto it = addInterval(is, L, R);
auto r2 = it->second;
if (it->first == L) is.erase(it);
else (int&)it->second = L;
if (R != r2) is.emplace(R, r2);
}
```

2.10 Mo's on Tree

$ST(u) \leq ST(v)$
 $P = LCA(u, v)$
If $P = u$, query $[ST(u), ST(v)]$
Else query $[EN(u), ST(v)] + [ST(P), ST(P)]$

2.11 DSU Rollback

```
struct RollbackUF{
    vi e;
    vector<pii> st;
    RollbackUF(int n) : e(n, -1){}
    int size(int x){
        return -e[find(x)];
    }
    int find(int x){
        return (e[x] < 0) ? x : find(e[x]);
    }
    int time(){
        return st.size();
    }
    void rollback(int t){
        for (int i = time(); i-->t; )
            e[st[i].fi] = st[i].se;
        st.resize(t);
    }
    bool join(int a, int b){
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
        st.push_back({a, e[a]});
        st.push_back({b, e[b]});
        e[a] += e[b];
        e[b] = a;
        return true;
    }
};
```

2.12 Link-Cut Tree

```
// sz for path queries
// sub, vsub for subtree queries
// x->access() brings x to the top and propagates it,
// x's left subtree is path from x to root
// after access, sub is no. of nodes in CC of x,
// vsub is no. of nodes under x
// use makeRoot() for arbitrary path queries

typedef struct Node* Np;
struct Node {
    Np p, c[2]; // parent, children
    bool flip = 0; // subtree flipped or not
    int vtx;
    ll val, sz; // value in node, # nodes in ←
                // current splay tree
    int sub, vsub = 0; // # of nodes in CC
    ll stsum; // sum of all val in the splay ←
                // tree
    // after access(): from original root to this node
    Node(int _vtx, int _val = 1) : vtx(_vtx), val(_val) {
        p = c[0] = c[1] = nullptr;
        calc();
    }

    friend int getSz(Np x) {
        return x ? x->sz : 0;
    }
    friend int getSub(Np x) {
        return x ? x->sub : 0;
    }
    friend ll getStSum(Np x) {
        return x ? x->stsum : 0;
    }

    void prop() { // lazy prop
        if(!flip)
            return;
```

```
        swap(c[0], c[1]);
        flip = 0;
        for(int i = 0; i < 2; i++) {
            if(c[i])
                c[i]->flip ^= 1;
        }
    }
    void calc() { // recalc vals
        for(int i = 0; i < 2; i++) {
            if(c[i])
                c[i]->prop();
        }
        sz = 1 + getSz(c[0]) + getSz(c[1]);
        sub = 1 + getSub(c[0]) + getSub(c[1]) + vsub;
        stsum = val + getStSum(c[0]) + getStSum(c[1]);
    }

    // SPLAY TREE OPERATIONS
    int dir() { // p is path-parent pointer
        if(!p)
            return -2;
        for(int i = 0; i < 2; i++) {
            if(p->c[i] == this)
                return i;
        }
        return -1; // -> not in current splay tree
    }
    // test if root of current splay tree
    bool isRoot() {
        return dir() < 0;
    }
    friend void setLink(Np x, Np y, int d) { // x is orig←
        parent
        if(y)
            y->p = x;
        if(d >= 0)
            x->c[d] = y;
    }
    void rot() { // assume p and p->p propagated
        assert(!isRoot());
        int x = dir();
        Np pa = p;
        setLink(pa->p, this, pa->dir());
        setLink(pa, c[x ^ 1], x);
        setLink(this, pa, x ^ 1);
        pa->calc();
    }
    void splay() { // bring this node to the root of ←
        splay tree
        while(!isRoot() && !p->isRoot()) {
            p->p->prop(), p->prop(), prop();
            dir() == p->dir() ? p->rot() : rot();
            rot();
        }
        if(!isRoot())
            p->prop(), prop(), rot();
        prop();
        calc();
    }
    Np fbo(int b) { // find by order
        prop();
        int z = getSz(c[0]); // of splay tree
        if(b == z) {
            splay();
            return this;
        }
        return b < z ? c[0]->fbo(b) : c[1]->fbo(b - z - 1);
    }

    // BASE OPERATIONS
```

```
// bring this to top of splay tree (not impacting ←
    // original tree)
void access() {
    for(Np v = this, pre = nullptr; v; v = v->p) {
        v->splay(); // now switch virtual children
        if(pre)
            v->vsub -= pre->sub;
        if(v->c[1])
            v->vsub += v->c[1]->sub;
        v->c[1] = pre;
        v->calc();
        pre = v;
    }
    splay();
    assert(!c[1]); // right subtree is empty
}
void makeRoot() { // of the original tree
    access();
    flip ^= 1;
    access();
    assert(!c[0] && !c[1]);
}

// QUERIES
friend Np lca(Np x, Np y) {
    if(x == y)
        return x;
    x->access(), y->access();
    if(!x->p)
        return nullptr;
    x->splay();
    return x->p ? : x; // y was below x in latter case
    // access at y did not affect x←
    -> not connected
    friend bool connected(Np x, Np y) {
        return lca(x, y);
    }

    // no. of nodes above; distance to root in original ←
    // tree
    int distRoot() {
        access();
        return getSz(c[0]);
    }
    Np getRoot() { // get root of LCT component in ←
        original tree
        access();
        Np a = this;
        while(a->c[0])
            a = a->c[0], a->prop();
        a->access();
        return a;
    }
    Np getPar(int b) { // get b-th parent on path to root
        access();
        b = getSz(c[0]) - b;
        assert(b >= 0);
        return fbo(b);
    } // can also get min, max on path to root, etc

    // MODIFICATIONS
    void setVal(int v) {
        access();
        val = v;
        calc();
    }
    void addVal(int v) {
        access();
        val += v;
        calc();
    }
}
```

```
friend void link(Np x, Np y, bool force = 1) {
    assert(!connected(x, y));
    if(force) {
        y->makeRoot(); /// make x par of y; x -> y
    } else {
        y->access();
        assert(!y->c[0]);
    }
    x->access();
    setLink(y, x, 0);
    y->calc();
}
friend void cut(Np y) { // cut y from its parent
    y->access();
    assert(y->c[0]);
    y->c[0]->p = NULL;
    y->c[0] = NULL;
    y->calc();
}
friend void cut(Np x, Np y) { // if x, y adj in tree
    x->makeRoot();
    y->access();
    assert(y->c[0] == x && !x->c[0] && !x->c[1]);
    cut(y);
}
};
Np LCT[maxV];
```

2.13 Max Queue

```
//For min queue, change every Max to Min, and < to >
struct MaxQueue{
    deque<pair<int,int>> q;
    int cntAdd, cntRem;

    MaxQueue() : cntAdd(0), cntRem(0) {};

    void add(int x){
        while(!q.empty()&&q.back().first<x)q.pop_back();
        q.push_back({x,cntAdd++});
    }
    void remove(){
        if (!q.empty() && q.front().second == cntRem)
            q.pop_front();
        cntRem++;
    }
    int getMax(){
        return q.front().first;
    }
};
```

3 Dynamic Programming

3.1 DP Convex Hull

```
/* dp[i] = min k<i {dp[k] + x[i]*m[k]}
   Make sure gradient (m[i]) is either non-increasing ←
   if min,
   or non-decreasing if max. x[i] must be non-←
   decreasing. just sort */
int y[N], m[N];
// while this is true, pop back from dq. a=new line, b=←
last, c=2nd last
bool cekx(int a, int b, int c) {
    // if not enough, change to cross mul
    // if cross mul, beware of negative denominator, and ←
    overflow
```

```
return (double)(y[b] - y[a]) / (m[a] - m[b]) <= (←
double)(y[c] - y[b]) /
(m[b] - m[c]);
}
```

3.2 DP DNC

```
void f(int rem, int l, int r, int optl, int optr) {
    if(l > r)
        return;
    int mid = l + r >> 1;
    int opt = MOD, optid = mid;
    for(int i = optl; i <= mid && i <= optr; ++i) {
        if(dp[rem - 1][i] + c[i][mid] < opt) {
            opt = dp[rem - 1][i] + c[i][mid];
            optid = i;
        }
    }
    dp[rem][mid] = opt;
    f(rem, l, mid - 1, optl, optid);
    f(rem, mid + 1, r, optid, optr);
    return;
}
rep(i, 1, n)dp[1][i] = c[0][i];
rep(i, 2, k)f(i, i, n, i, n);
```

3.3 DP Knuth-Yao

```
// opt[i+1][j] <= opt[i][j] <= opt[i][j+1]
// dp[i][j] = min{k} dp[i][k]+dp[k][j]+cost[i][j]
for(int k = 0; k <= n; k++) {
    for(int i = 0; i + k <= n; i++) {
        if(k < 2)
            dp[i][i + k] = 0, opt[i][i + k] = i;
        else {
            int sta = opt[i][i + k - 1];
            int end = opt[i + 1][i + k];
            for(int j = sta; j <= end; j++) {
                if(dp[i][j] + dp[j][i + k] + cost[i][i + k] < ←
                dp[i][i + k]) {
                    dp[i][i + k] = dp[i][j] + dp[j][i + k] + cost←
                    [i][i + k];
                    opt[i][i + k] = j;
                }
            }
        }
    }
}
```

4 Geometry

4.1 Geometry Template

```
/*
TABLE OF CONTENT
0. Basic Rule
    0.1. Everything is in double
    0.2. Every comparison use EPS
    0.3. Every degree in rad
1. General Double Operation
    1.1. const double EPS=1E-9
    1.2. const double PI=acos(-1.0)
    1.3. const double INF=1E9
    1.3. between_d(double x,double l,double r)
        check whether x is between l and r inclusive ←
        with EPS
    1.4. same_d(double x,double y)
```

```
check whether x=y with EPS
1.5. dabs(double x)
    absolute value of x

2. Point
2.1. struct point
    2.1.1. double x,y
        cartesian coordinate of the point
    2.1.2. point()
        default constructor
    2.1.3. point(double _x,double _y)
        constructor, set the point to (_x,_y)
    2.1.4. bool operator< (point other)
        regular pair<double,double> operator < with←
        EPS
    2.1.5. bool operator== (point other)
        regular pair<double,double> operator == ←
        with EPS
2.2. hypot(point P)
    length of hypotenuse of point P to (0,0)
2.3. e_dist(point P1,point P2)
    euclidean distance from P1 to P2
2.4. m_dist(point P1,point P2)
    manhattan distance from P1 to P2
2.5. pointBetween(point P, point L, point R)
    check if point P is in between L and R
2.6. collinear(point P, point L, point R)
    check if the three points are collinear
2.7. mid(point P, point Q)
    find mid point of P and Q
```

```
3.Vector
3.1.structvec
    3.1.1. double x, y
        x and y are magnitudes of the vector
    3.1.2. vec()
        defaultconstructor
    3.1.3. vec(double _x, double _y)
        constructor, set the vector to (_x, _y)
    3.1.4. vec(point A, point B)
        constructor, set the vector to vector AB (A←
        ->B)
3.2. scale(vec v, double x)
    scale vector v by x
3.3. flip(vec v)
    flip the direction of v
3.4. dot(vec u, vec v)
3.5. cross(vec u, vec v)
3.6. norm_sq(vec v)
    return length of v

4. Additional Point Operators (needs point and vec)
4.1. rotate (point P, point O, double angle)
    rotate point P from the origin O by angle (CCW)
4.2. angle (point A, point O, point B)
    check the angle of AOB
4.3. orientation (point P, point Q, point R)
    check the orientation of PR towards PQ

5. Line
5.1. struct line
    5.1.1. double a, b, c
        coefficients of line in ax+by+c=0
    5.1.2. line()
        default constructor
    5.1.3. line (double _a, double _b, double _c)
        makeline_ax+_by+_c=0
    5.1.4. line (point P1, point P2)
        make line that goes through P1 and P2
    5.1.5. line (point P, double slope)
        make line that goes through P and has slope
```



```

5.1.6.bool operator==
5.1.7.double slope()
    return slope of the line
5.2. paralel (line L1, line L2)
    check if L1 and L2 are parallel
5.3. intersection (line L1, line L2, point &P)
    check if L1 and L2 intersects,
    and return the intersection point P
5.4. pointToLine (point P, point A, point B, point <-
    &C)
    return the distance of point P to line AB
    and the projection point C
    note: only needs point and vec
5.5. lineToLine (line L1, line L2)
    return the distance between L1 and L2

6. Line Segment (needs point and line)
6.1.structsegment
6.1.1. point P, Q
    start point and end point
6.1.2. line L -> line equation
6.1.3. segment()
    default constructor
6.1.4. segment (point _P, point _Q)
    make segment from point P to Q
6.1.5. bool operator==
6.2. onSegment (point P, segment S)
    check if point P is on segment S
6.3. s_intersection (segment S1, segment S2)
    check if two segments intersect
6.4. pointToSegment (point P, point A, point B, <-
    point &C)
    return the distance of point P to segment AB
    and the projection point C
6.5. segmentToSegment(segment S1, segment S2)
    return the distance of two segments
6.6. lineIntersectSeg (point p, point q, point A, <-
    point B)
    assuming they intersect, return the <-
    intersection point
    of segment pq and line AB\

7.Circle
7.1.structcircle
7.1.1. point P -> center point
7.1.2. double r -> radius
7.1.3. circle()
    default constructor
7.1.4. circle(point _P, double _r)
    make circle with center point P and radius <-
    r
7.1.5. circle (point P1, point P2)
    make circle with diameter P1P2
7.1.6. circle(point P1, point P2, point P3)
    make circle that pass through P1, P2, and <-
    P3
7.1.7. bool operator==
7.2. insideCircle (point P, circle C)
    check if point P is inside circle C
7.3. c_intersection (circle C1, circle C2, point &<-
    P1, point &P2)
    check if two circle intersects
    and return the points in P1 and P2
7.4. lc_intersection (line L, circle O, point &P1, <-
    point &P2)
    check if line and circle intersects
    and return the points in P1 and P2
7.5. sc_intersection (segment S, circle C, point &<-
    P1, point &P2)
    check if segment and circle intersects
    and return the points in P1 and P2

```

```

8.Triangle
8.1. t_perimeter (point A, point B, point C)
    return perimeter of triangle
8.2. t_area (point A, point B, point C)
    return area of triangle
8.3. t_inCircle (point A, point B, point C)
    return in-circle of the triangle
8.4. t_outCircle (point A, point B, point C)
    return circum-circle of the triangle

9.Polygon
9.1. struct polygon
9.1.1. vector<point> P
    list of points in the polygon
9.1.2. polygon()
    default constructor
9.1.3. polygon (const vector<point>& _P)
    make polygon from point list
9.2. rayCast (point P, polygon &A)
    check if point is inside polygon
9.3. cutPolygon(point a, point b, const vector<<-
    point> &Q)
    cuts polygon Q along the line formed
    by point a and point b
9.4. minCoverCircle (polygon &A)
    return circle that covers polygon

10. Other Geometry Algorithms
10.1. minCostPolygonTriangulation(polygon &A)
    return min cost of polygon triangulation

*/
//=====
//1.General Double Operations
//=====
#define For(i, a, b) for(int i = (a); i <= (b); ++i)
#define rep(i, a, b) for(int i = (a); i <= (b); ++i)
#define pb push_back

const double PI = acos(-1.0);
const double INF9 = 1E9;
const double EPS = 1e-9;
double between_d(double x, double l, double r){
    return (min(l,r) <= x + EPS && x <= max(l,r) + EPS);
}
double same_d (double x, double y){
    return between_d (x, y, y);
}
double dabs(double x){
    if (x < EPS) return -x;
    return x;
}
//=====
//2.Point
//=====
struct point{
    double x, y;
    point() {x = y = 0.0;}
    point(double _x, double _y) : x(_x), y(_y){}
    bool operator< (point other){
        if (x < other.x + EPS) return true;
        if(x + EPS > other.x) return false;
        return y < other.y + EPS;
    }
    bool operator== (point other){
        return same_d(x, other.x) && same_d(y, other.y);
    }
};
double e_dist (point P1, point P2){
    return hypot (P1.x - P2.x, P1.y - P2.y);
}

```

```

double m_dist(point P1, point P2){
    return dabs(P1.x - P2.x) + dabs(P1.y - P2.y);
}
double pointBetween(point P, point L, point R){
    return(e_dist(L,P) + e_dist(P,R) == e_dist(L,R));
}
bool collinear(point P, point L, point R){
    return dabs(P.x * (L.y - R.y) + L.x * (R.y - P.y)
    + R.x * (P.y - L.y)) < EPS;
}
point mid(point P, point Q){
    return point((P.x + Q.x)/2, (P.y + Q.y)/2);
}
//=====
//3.Vector
//=====
struct vec{
    double x, y;
    vec(){x = y = 0.0;}
    vec(double _x, double _y) : x(_x), y(_y){}
    vec(point A) : x(A.x), y(A.y){}
    vec(point A, point B) : x(B.x - A.x), y(B.y - A.y){}
};
vec scale(vec v, double s){
    return vec(v.x * s, v.y * s);
}
vec flip(vec v){
    return vec(-v.x, -v.y);
}
double dot(vec u, vec v){
    return(u.x * v.x + u.y * v.y);
}
double cross(vec u, vec v){
    return(u.x * v.y - u.y * v.x);
}
double norm_sq(vec v){
    return(v.x * v.x + v.y * v.y);
}
//=====
//4.AdditionalPointOperators
//=====
point translate(point P, vec v){
    return point(P.x + v.x, P.y + v.y);
}
point rotate(point P, point O, double angle){
    vec v(O);
    P = translate(P, flip(v));
    return translate(point(P.x * cos(angle) - P.y * sin(<-
    angle),
    P.x * sin(angle) + P.y * cos(angle)), v);
}
double angle(point A, point O, point B){
    vec OA (O,A), OB (O,B);
    return acos(dot(OA,OB) / sqrt(norm_sq(OA) * norm_sq(<-
    OB)));
}
int orientation(point P, point Q, point R){
    vec PQ(P,Q), PR(P,R);
    double c = cross(PQ, PR);
    if (c < -EPS) return -1;
    if(c > EPS) return 1;
    return 0;
}
//=====
//5.Line
//=====
struct line{
    double a, b, c;
    line(){a = b = c = 0.0;}
    line(double _a, double _b, double _c) : a(_a), b(_b),<-
    c(_c){}
}

```

```

line(point P1, point P2){
    if (P1 < P2) swap(P1, P2);
    if (same_d(P1.x, P2.x))
        a = 1.0, b = 0.0, c = -P1.x;
    else
        a = -(P1.y - P2.y)/(P1.x - P2.x),
        b = 1.0, c = -(a * P1.x) - P1.y;
}
line(point P, double slope){
    if (same_d(slope, INFD))
        a = 1.0, b = 0.0, c = -P.x;
    else
        a = -slope, b = 1.0, c = -(a * P.x) - P.y;
}
bool operator==(line other){
    return same_d(a, other.a) && same_d(b, other.b)
    && same_d(c, other.c);
}
double slope(){
    if (same_d(b, 0.0))return INFD;
    return -(a/b);
}
};
bool paralel(line L1, line L2){
    return same_d(L1.a, L2.a) && same_d(L1.b, L2.b);
}
bool intersection(line L1, line L2, point &P){
    if (paralel(L1, L2)) return false;
    P.x = (L2.b*L1.c - L1.b*L2.c) / (L2.a*L1.b - L1.a*L2.c);
    b);
    if (same_d(L1.b, 0.0))
        P.y = -(L2.a * P.x + L2.c);
    else
        P.y = -(L1.a * P.x + L1.c);
    return true;
}
double pointToLine(point P, point A, point B, point &C){
    {
        vec AP(A,P), AB(A,B);
        double u = dot(AP,AB) / norm_sq(AB);
        C = translate(A, scale(AB, u));
        return e_dist(P, C);
    }
}
double lineToLine(line L1, line L2){
    if (!paralel(L1,L2)) return 0.0;
    return dabs(L2.c - L1.c) / sqrt(L1.a * L1.a + L1.b * L1.b);
}
//=====
//6.LineSegment
//=====
struct segment{
    point P, Q;
    line L;
    segment(){
        point T1;
        P = Q = T1;
        line T2;
        L = T2;
    }
    segment(point _P, point _Q){
        P = _P;
        Q = _Q;
        if (Q < P) swap(P, Q);
        line T(P, Q);
        L = T;
    }
    bool operator== (segment other){
        return P == other.P && Q == other.Q;
    }
};

```

```

bool onSegment(point P, segment S){
    if (orientation(S.P, S.Q, P) != 0)
        return false;
    return between_d(P.x, S.P.x, S.Q.x)
    && between_d(P.y, S.P.y, S.Q.y);
}
bool s_intersection(segment S1, segment S2){
    double o1 = orientation(S1.P, S1.Q, S2.P);
    double o2 = orientation(S1.P, S1.Q, S2.Q);
    double o3 = orientation(S2.P, S2.Q, S1.P);
    double o4 = orientation(S2.P, S2.Q, S1.Q);
    if (o1 != o2 && o3 != o4)
        return true;
    if (o1 == 0 && onSegment(S2.P, S1))
        return true;
    if (o2 == 0 && onSegment(S2.Q, S1))
        return true;
    if (o3 == 0 && onSegment(S1.P, S2))
        return true;
    if (o4 == 0 && onSegment(S1.Q, S2))
        return true;
    return false;
}
double pointToSegment(point P, point A, point B, point &C){
    &C){
        vec AP(A, P), AB(A, B);
        double u = dot(AP, AB) / norm_sq(AB);
        if (u < EPS){
            C = A;
            return e_dist(P, A);
        }
        if (u + EPS > 1.0){
            C = B;
            return e_dist(P, B);
        }
        return pointToLine(P, A, B, C);
    }
}
double segmentToSegment(segment S1, segment S2){
    if (s_intersection(S1, S2)) return 0.0;
    double ret = INFD;
    point dummy;
    ret = min({ret,
        pointToSegment(S1.P, S2.P, S2.Q, dummy),
        pointToSegment(S1.Q, S2.P, S2.Q, dummy),
        pointToSegment(S2.P, S1.P, S1.Q, dummy),
        pointToSegment(S2.Q, S1.P, S1.Q, dummy)
    });
    return ret;
}
//line segment p-q intersect with line A-B.
//note: Tak boleh berimpitan
point lineIntersectSeg(point p, point q, point A, point &B){
    B){
        double a = B.y - A.y;
        double b = A.x - B.x;
        double c = B.x * A.y - A.x * B.y;
        double u = fabs(a * p.x + b * p.y + c);
        double v = fabs(a * q.x + b * q.y + c);
        return point((p.x * v + q.x * u) / (u+v),
            (p.y * v + q.y * u) / (u+v));
    }
}
//=====
//7.Circle
//=====
struct circle{
    point P;
    double r;
    circle() : P(point()), r(0.0){}
    circle(point _P, double _r) : P(_P), r(_r){}
    circle(point P1, point P2) : P(mid(P1, P2)), r(e_dist(
        P, P1)){
}

```

```

circle(point P1, point P2, point P3){
    vector<point> T;
    T.clear();
    T.pb(P1);
    T.pb(P2);
    T.pb(P3);
    sort(T.begin(),T.end());
    P1 = T[0];
    P2 = T[1];
    P3 = T[2];
    point M1, M2;
    M1 = mid(P1, P2);
    M2 = mid(P2, P3);
    point Q2, Q3;
    Q2 = rotate(P2, P1, PI/2);
    Q3 = rotate(P3, P2, PI/2);
    vec P1Q2(P1,Q2), P2Q3(P2,Q3);
    point M3, M4;
    M3 = translate(M1, P1Q2);
    M4 = translate(M2, P2Q3);
    line L1(M1, M3), L2(M2, M4);
    intersection(L1, L2, P);
    r = e_dist(P,P1);
}
bool operator== (circle other){
    return (P == other.P && same_d(r, other.r));
}
};
bool insideCircle(point P, circle C){
    return e_dist(P, C.P) <= C.r + EPS;
}
bool c_intersection(circle C1, circle C2, point &P1, &P2){
    point &P2){
        double d = e_dist(C1.P, C2.P);
        if (d > C1.r + C2.r){
            return false; //d+EPS kalo butuh
        }
        if (d < dabs(C1.r - C2.r) + EPS) return false;
        double x1 = C1.P.x, y1 = C1.P.y, r1 = C1.r,
            x2 = C2.P.x, y2 = C2.P.y, r2 = C2.r;
        double a = (r1*r1 - r2*r2 + d*d)/(2*d),
            h = sqrt(r1*r1 - a*a);
        point T(x1 + a*(x2 - x1)/d,
            y1 + a*(y2 - y1)/d);
        P1 = point(T.x - h*(y2-y1)/d,
            T.y + h*(x2-x1)/d);
        P2 = point(T.x + h*(y2-y1)/d,
            T.y - h*(x2-x1)/d);
        return true;
    }
}
bool lc_intersection(line L, circle O, point &P1, point &P2){
    &P2){
        double a = L.a, b = L.b, c = L.c,
            x = O.P.x, y=O.P.y, r=O.r;
        double A = a*a + b*b,
            B = 2*a*b*x - 2*a*c - 2*b*b*x,
            C = b*b*x*x + b*b*y*y - 2*b*c*y+c*c - b*b*r*r;
        double D = B*B - 4*A*C;
        point T1, T2;
        if (same_d(b, 0.0)){
            T1.x = c/a;
            if (dabs(x - T1.x) + EPS > r)
                return false;
            if (same_d(T1.x - r - x, 0.0) || same_d(T1.x + r - x, 0.0)){
                P1 = P2 = point(T1.x, y);
                return true;
            }
        }
        double dx = dabs(T1.x - x), dy = sqrt(r*r - dx*dx);
        P1 = point(T1.x, y - dy);
        P2 = point(T1.x, y + dy);
    }
}

```



```

        return true;
    }
    if (same_d(D, 0.0)){
        T1.x = -B / (2*A);
        T1.y = (c - a*T1.x) / b;
        P1 = P2 = T1;
        return true;
    }
    if (D < EPS) return false;
    D = sqrt(D);
    T1.x = (-B - D) / (2*A);
    T1.y = (c - a*T1.x) / b;
    P1 = T1;
    T2.x = (-B + D) / (2*A);
    T2.y = (c - a*T2.x) / b;
    P2 = T2;
    return true;
}

bool sc_intersection(segment S, circle C, point &P1, ←
    point &P2){
    bool cek = lc_intersection(S.L, C, P1, P2);
    if (!cek) return false;
    double x1 = S.P.x, y1 = S.P.y,
           x2 = S.Q.x, y2 = S.Q.y;
    bool b1 = between_d(P1.x, x1, x2) && between_d(P1.y, ←
        y1, y2);
    bool b2 = between_d(P2.x, x1, x2) && between_d(P2.y, ←
        y1, y2);
    if (P1 == P2) return b1;
    if (b1 || b2){
        if (!b1) P1 = P2;
        if (!b2) P2 = P1;
        return true;
    }
    return false;
}

//=====
//8.Triangle
//=====
double t_perimeter(point A, point B, point C){
    return e_dist(A,B) + e_dist(B,C) + e_dist(C,A);
}

double t_area(point A, point B, point C){
    double s = t_perimeter(A,B,C)/2;
    double ab = e_dist(A,B), bc = e_dist(B,C), ac = ←
        e_dist(C,A);
    return sqrt(s*(s-ab)*(s-bc)*(s-ac));
}

circle t_inCircle(point A, point B, point C){
    vector<point> T;
    T.clear();
    T.pb(A);
    T.pb(B);
    T.pb(C);
    sort(T.begin(),T.end());
    A = T[0];
    B = T[1];
    C = T[2];
    double r = t_area(A,B,C) / (t_perimeter(A,B,C)/2);
    double ratio = e_dist(A,B) / e_dist(A,C);
    vec BC(B,C);
    BC = scale(BC, ratio/(1 + ratio));
    point P;
    P = translate(B, BC);
    line AP1(A, P);
    ratio = e_dist(B,A)/e_dist(B,C);
    vec AC(A,C);
    AC = scale(AC,ratio / (1+ratio));
    P = translate(A, AC);
    line BP2(B, P);
    intersection(AP1, BP2, P);

```

```

        return circle(P, r);
    }
    circle t_outCircle(point A, point B, point C){
        return circle(A,B,C);
    }
    //=====
    //9.Polygon
    //=====
    struct polygon{
        vector<point> P;
        polygon(){P.clear();}
        polygon(const vector<point>& _P) : P(_P){}
    };
    bool rayCast(point P, polygon &A) {
        bool inside = false;
        int n = A.P.size();
        for (int i = 0, j = n - 1; i < n; j = i++) {
            if (((A.P[i].y > P.y) != (A.P[j].y > P.y)) &&
                (P.x < (A.P[j].x - A.P[i].x) * (P.y - A.P[i].y) / ←
                    (A.P[j].y - A.P[i].y) + A.P[i].x))
                inside = !inside;
        }
        return inside;
    }
    // (note: the last point must be the same as the first ←
    point)
    vector<point> cutPolygon(point a, point b, const vector<←
        point> &Q){
        vector<point> P;
        for (int i = 0; i < (int)Q.size(); i++){
            double left1 = cross(vec(a, b), vec(a, Q[i]));
            double left2 = 0;
            if (i != (int)Q.size() - 1)
                left2 = cross(vec(a,b), vec(a, Q[i+1]));
            if (left1 > -EPS)
                P.push_back(Q[i]);
            if (left1 * left2 < -EPS)
                P.push_back(lineIntersectSeg(Q[i], Q[i+1], a, b))←
                    ;
        }
        if (!P.empty() && !(P.back() == P.front()))
            P.push_back(P.front());
        return P;
    }
    circle minCoverCircle(polygon &A){
        vector<point> p = A.P;
        point c;
        circle ret;
        double cr = 0.0;
        int i, j, k;
        c = p[0];
        for (i = 1; i < p.size(); i++){
            if (e_dist(p[i], c) >= cr + EPS){
                c = p[i], cr = 0;
                ret = circle(c, cr);
                for (j = 0; j < i; j++){
                    if (e_dist(p[j], c) >= cr + EPS){
                        c = mid(p[i], p[j]);
                        cr = e_dist(p[i], c);
                        ret = circle(c, cr);
                        for (k = 0; k < j; k++){
                            if (e_dist(p[k], c) >= cr + EPS){
                                ret = circle(p[i], p[j], p[k]);
                                c = ret.P;
                                cr = ret.r;
                            }
                        }
                    }
                }
            }
        }
    }
}

```

```

        return ret;
    }
    //=====
    //10. Other Geometry Algorithm
    //=====
    double DP[110][110];
    double minCostPolygonTriangulation(polygon &A){
        if (A.P.size() < 3) return 0;
        For(i, 0, A.P.size()){
            for(int j = 0, k = i; k < A.P.size(); j++, k++){
                if (k < j+2)
                    DP[j][k] = 0.0;
                else{
                    DP[j][k] = INF;
                    rep(l, j+1, k-1){
                        double cost = e_dist(A.P[j], A.P[k])
                            + e_dist(A.P[k], A.P[l])
                            + e_dist(A.P[l], A.P[j]);
                        DP[j][k] = min(DP[j][k], DP[j][l] + DP[l][k←
                            ] + cost);
                    }
                }
            }
        }
        return DP[0][A.P.size() - 1];
    }
}

```

4.2 Convex Hull

```

typedef double TD; // for precision ←
                shifts
namespace GEOM {
    typedef pair<TD, TD> Pt; // vector and points
    const TD EPS = 1e-9;
    const TD maxD = 1e9;
    TD cross(Pt a, Pt b, Pt c) { // right hand rule
        TD v1 = a.first - c.first; // (a-c) X (b-c)
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u2 - v2 * u1;
    }
    TD cross(Pt a, Pt b) { // a X b
        return a.first * b.second - a.second * b.first;
    }
    TD dot(Pt a, Pt b, Pt c) { // (a-c) . (b-c)
        TD v1 = a.first - c.first;
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u1 + v2 * u2;
    }
    TD dot(Pt a, Pt b) { // a . b
        return a.first * b.first + a.second * b.second;
    }
    TD dist(Pt a, Pt b) {
        return sqrt((a.first - b.first) * (a.first - b.←
            first) +
            (a.second - b.second) * (a.second - b.←
            second));
    }
    TD shoelaceX2(vector<Pt>& convHull) {
        TD ret = 0;
        for(int i = 0, n = convHull.size(); i < n; i++)
            ret += cross(convHull[i], convHull[(i + 1) % n]);
        return ret;
    }
    vector<Pt> createConvexHull(vector<Pt>& points) {
        sort(points.begin(), points.end());
        vector<Pt> ret;
    }
}

```

```
for(int i = 0; i < points.size(); i++) {
    while(ret.size() > 1 &&
        cross(points[i], ret[ret.size() - 1], ret[←
            ret.size() - 2]) < -EPS)
        ret.pop_back();
    ret.push_back(points[i]);
}
for(int i = points.size() - 2, sz = ret.size(); i ←
    >= 0; i--) {
    while(ret.size() > sz &&
        cross(points[i], ret[ret.size() - 1], ret[←
            ret.size() - 2]) < -EPS)
        ret.pop_back();
    if(i == 0)
        break;
    ret.push_back(points[i]);
}
return ret;
bool isInside(Pt pv, vector<Pt>& x) { //using ←
    winding number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first ←
            >= pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first ←
            <= pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].←
            second >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].←
            second <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
}
for(int i = 0; i < n; ++i) {
    if(x[i].second <= pv.second) {
        if(x[i + 1].second > pv.second && cross(x[i],←
            x[i + 1], pv) > 0)
            ++wn;
        } else if(x[i + 1].second <= pv.second && cross←
            (x[i], x[i + 1], pv) < 0)
            --wn;
    }
    x.pop_back();
    return wn != 0;
}
}
bool isInside(Pt pv, vector<Pt>& x) { //using winding←
    number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first >= ←
            pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first <= ←
            pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].second←
            >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].second←
            <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
}
for(int i = 0; i < n; ++i) {
    if(x[i].second <= pv.second) {
```

```
        if(x[i + 1].second > pv.second && cross(x[i], x←
            [i + 1], pv) > 0)
            ++wn;
        } else if(x[i + 1].second <= pv.second && cross(x←
            [i], x[i + 1], pv) < 0)
            --wn;
    }
    x.pop_back();
    return wn != 0;
}
}
```

4.3 Closest Pair of Points

```
#define fi first
#define se second
typedef pair<int, int> pii;
struct Point {
    int x, y, id;
};
int compareX(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->x - p2->x);
}
int compareY(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->y - p2->y);
}
double dist(Point p1, Point p2) {
    return sqrt((double)(p1.x - p2.x) * (p1.x - p2.x) +
        (double)(p1.y - p2.y) * (p1.y - p2.y)
    );
}
pair<pii, double> bruteForce(Point P[], int n) {
    double min = 1e8;
    pii ret = pii(-1, -1);
    for(int i = 0; i < n; ++i)
        for(int j = i + 1; j < n; ++j)
            if(dist(P[i], P[j]) < min) {
                ret = pii(P[i].id, P[j].id);
                min = dist(P[i], P[j]);
            }
    return pair<pii, double> (ret, min);
}
pair<pii, double> getmin(pair<pii, double> x, pair<pii,←
    double> y) {
    if(x.fi.fi == -1 && x.fi.se == -1)
        return y;
    if(y.fi.fi == -1 && y.fi.se == -1)
        return x;
    return (x.se < y.se) ? x : y;
}
pair<pii, double> stripClosest(Point strip[], int size,←
    double d) {
    double min = d;
    pii ret = pii(-1, -1);
    qsort(strip, size, sizeof(Point), compareY);
    for(int i = 0; i < size; ++i)
        for(int j = i + 1; j < size && (strip[j].y - strip[←
            i].y) < min; ++j)
            if(dist(strip[i], strip[j]) < min) {
                ret = pii(strip[i].id, strip[j].id);
                min = dist(strip[i], strip[j]);
            }
    return pair<pii, double>(ret, min);
}
pair<pii, double> closestUtil(Point P[], int n) {
    if(n <= 3)
        return bruteForce(P, n);
    int mid = n / 2;
```

```
    Point midPoint = P[mid];
    pair<pii, double> dl = closestUtil(P, mid);
    pair<pii, double> dr = closestUtil(P + mid, n - mid);
    pair<pii, double> d = getmin(dl, dr);
    Point strip[n];
    int j = 0;
    for(int i = 0; i < n; i++)
        if(abs(P[i].x - midPoint.x) < d.second)
            strip[j] = P[i], j++;
    return getmin(d, stripClosest(strip, j, d.second));
}
pair<pii, double> closest(Point P[], int n) {
    qsort(P, n, sizeof(Point), compareX);
    return closestUtil(P, n);
}
Point P[50005];
int main() {
    int n;
    scanf("%d", &n);
    for(int a = 0; a < n; a++) {
        scanf("%d%d", &P[a].x, &P[a].y);
        P[a].id = a;
    }
    pair<pii, double> hasil = closest(P, n);
    if(hasil.fi.fi > hasil.fi.se)
        swap(hasil.fi.fi, hasil.fi.se);
    printf("%d %d %.6lf\n", hasil.fi.fi, hasil.fi.se, ←
        hasil.se);
    return 0;
}
```

4.4 Smallest Enclosing Circle

```
// welzl's algo to find the 2d minimum enclosing circle←
    of a set of points
// expected O(N)
// directions: remove duplicates and shuffle points, ←
    then call welzl(points)

struct Point {
    double x;
    double y;
};

struct Circle {
    double x, y, r;
    Circle() {}
    Circle(double _x, double _y, double _r): x(_x), y(_y)←
        , r(_r) {}
};

Circle trivial(const vector<Point>& r) {
    if(r.size() == 0)
        return Circle(0, 0, -1);
    else if(r.size() == 1)
        return Circle(r[0].x, r[0].y, 0);
    else if(r.size() == 2) {
        double cx = (r[0].x + r[1].x) / 2.0, cy = (r[0].y + ←
            r[1].y) / 2.0;
        double rad = hypot(r[0].x - r[1].x, r[0].y - r[1].y)←
            ) / 2.0;
        return Circle(cx, cy, rad);
    } else {
        double x0 = r[0].x, x1 = r[1].x, x2 = r[2].x;
        double y0 = r[0].y, y1 = r[1].y, y2 = r[2].y;
        double d = (x0 - x2) * (y1 - y2) - (x1 - x2) * (y0 ←
            - y2);
        double cx = (((x0 - x2) * (x0 + x2) + (y0 - y2) * (←
            y0 + y2)) / 2 *
```

```

        (y1 - y2) - ((x1 - x2) * (x1 + x2) + (←
            y1 - y2) * (y1 + y2)) / 2
        * (y0 - y2)) / d;
    double cy = (((x1 - x2) * (x1 + x2) + (y1 - y2) * (←
        y1 + y2)) / 2 *
        (x0 - x2) - ((x0 - x2) * (x0 + x2) + (←
            y0 - y2) * (y0 + y2)) / 2
        * (x1 - x2)) / d;
    return Circle(cx, cy, hypot(x0 - cx, y0 - cy));
}
}

// SHUFFLE THE POINTS FIRST!!!!!!
Circle welzl(const vector<Point>& p, int idx = 0, ←
    vector<Point> r = {}) {
    if(idx == (int) p.size() || r.size() == 3)
        return trivial(r);
    Circle d = welzl(p, idx + 1, r);
    if(hypot(p[idx].x - d.x, p[idx].y - d.y) > d.r) {
        r.push_back(p[idx]);
        d = welzl(p, idx + 1, r);
    }
    return d;
}
}

```

4.5 Sutherland-Hodgman Algorithm

```

// Complexity: linear time
// Ada 2 poligon, cari poligon intersectionnya
// poly_point = hasilnya, clipper = pemotongnya
#include<bits/stdc++.h>
using namespace std;

const double EPS = 1e-9;

struct point {
    double x, y;
    point(double _x, double _y): x(_x), y(_y) {}
};
struct vec {
    double x, y;
    vec(double _x, double _y): x(_x), y(_y) {}
};

point pivot(0, 0);
vec toVec(point a, point b) {
    return vec(b.x - a.x, b.y - a.y);
}

double dist(point a, point b) {
    return hypot(a.x - b.x, a.y - b.y);
}

double cross(vec a, vec b) {
    return a.x * b.y - a.y * b.x;
}

bool ccw(point p, point q, point r) {
    return cross(toVec(p, q), toVec(p, r)) > 0;
}

bool collinear(point p, point q, point r) {
    return fabs(cross(toVec(p, q), toVec(p, r))) < EPS;
}

bool lies(point a, point b, point c) {
    if((c.x >= min(a.x, b.x) && c.x <= max(a.x, b.x)) &&
        (c.y >= min(a.y, b.y) && c.y <= max(a.y, b.y)))
        return true;
    else
        return false;
}

bool anglecmp(point a, point b) {
    if(collinear(pivot, a, b))
        return dist(pivot, a) < dist(pivot, b);
}

```

```

double d1x = a.x - pivot.x, d1y = a.y - pivot.y;
double d2x = b.x - pivot.x, d2y = b.y - pivot.y;
return (atan2(d1y, d1x) - atan2(d2y, d2x)) < 0;
}

point intersect(point s1, point e1, point s2, point e2)←
{
    double x1, x2, x3, x4, y1, y2, y3, y4;
    x1 = s1.x;
    y1 = s1.y;
    x2 = e1.x;
    y2 = e1.y;
    x3 = s2.x;
    y3 = s2.y;
    x4 = e2.x;
    y4 = e2.y;
    double num1 = (x1 * y2 - y1 * x2) * (x3 - x4) - (x1 - ←
        x2) * (x3 * y4 - y3 * x4);
    double num2 = (x1 * y2 - y1 * x2) * (y3 - y4) - (y1 - ←
        y2) * (x3 * y4 - y3 * x4);
    double den = (x1 - x2) * (y3 - y4) - (y1 - y2) * (x3 - ←
        x4);
    double new_x = num1 / den;
    double new_y = num2 / den;
    return point(new_x, new_y);
}

void clip(vector<point>& poly_points, point point1, ←
    point point2) {
    vector<point> new_points;
    new_points.clear();
    for(int i = 0; i < poly_points.size(); i++) {
        int k = (i + 1) % poly_points.size();
        double i_pos = ccw(point1, point2, poly_points[i]);
        double k_pos = ccw(point1, point2, poly_points[k]);
        //in in
        if(i_pos <= 0 && k_pos <= 0)
            new_points.push_back(poly_points[k]);
        //out in
        else if(i_pos > 0 && k_pos <= 0) {
            new_points.push_back(intersect(point1, point2, ←
                poly_points[i],
                poly_points[k]));
            new_points.push_back(poly_points[k]);
        }
        // in out
        else if(i_pos <= 0 && k_pos > 0) {
            new_points.push_back(intersect(point1, point2, ←
                poly_points[i],
                poly_points[k]));
        }
        //out out
        else {
        }
    }
    poly_points.clear();
    for(int i = 0; i < new_points.size(); i++)
        poly_points.push_back(new_points[i]);
}

double area(const vector<point>& P) {
    double result = 0.0;
    double x1, y1, x2, y2;
    for(int i = 0; i < P.size() - 1; i++) {
        x1 = P[i].x;
        y1 = P[i].y;
        x2 = P[i + 1].x;
        y2 = P[i + 1].y;
        result += (x1 * y2 - x2 * y1);
    }
    return fabs(result) / 2;
}
}

```

```

void suthHodgClip(vector<point>& poly_points, vector<←
    point> clipper_points) {
    for(int i = 0; i < clipper_points.size(); i++) {
        int k = (i + 1) % clipper_points.size();
        clip(poly_points, clipper_points[i], clipper_points←
            [k]);
    }
}

vector<point> sortku(vector<point> P) {
    int P0 = 0;
    int i;
    for(i = 1; i < 3; i++) {
        if(P[i].y < P[P0].y || (P[i].y == P[P0].y && P[i].x←
            > P[P0].x))
            P0 = i;
    }
    point temp = P[0];
    P[0] = P[P0];
    P[P0] = temp;
    pivot = P[0];
    sort(++P.begin(), P.end(), anglecmp);
    reverse(++P.begin(), P.end());
    return P;
}

int main {
    clipper_points = sortku(clipper_points);
    suthHodgClip(poly_points, clipper_points);
}

```

4.6 Centroid of Polygon

$$C_x = \frac{1}{6A} \sum_{i=0}^{n-1} (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

$$C_y = \frac{1}{6A} \sum_{i=0}^{n-1} (y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

4.7 Pick Theorem

A: Area of a simply closed lattice polygon
 B: Number of lattice points on the edges
 I: Number of points in the interior
 $A = I + \frac{B}{2} - 1$

5 String

5.1 KMP

```

//s = string, t = pattern
//result is in 0-index (points to the index that ←
    matches)
int lps[t.size() + 1];
inline void precalc(string t){
    lps[0] = -1;
    for (int i = 1; i < (int)t.size(); i++){
        int j = lps[i-1];
        while (j >= 0 && t[i] != t[j+1]) j = lps[j];
        lps[i] = (t[i] == t[j+1]) ? (j+1) : -1;
    }
}

bool kmp(string s){
    for (int i = 0, j = -1; i < (int)s.size(); i++){
        while (j >= 0 && s[i] != t[j+1]) j = lps[j];
        if (s[i] == t[j+1]) j++;
        if (j == (int)t.size()-1) return 1;
    }
    return 0;
}
}

```

5.2 Aho-Corasick

```
const int K=26;
struct Vertex{
    int next[K];
    bool leaf = 0;
    int p = -1, ans = 0;
    char pch;
    int link = -1,mlink = -1;
    //magic link, is the link to find the nearest leaf
    int go[K];
    Vertex(int p = -1, char ch = '$') : p(p), pch(ch){
        fill(begin(next), end(next), -1);
        fill(begin(go), end(go), -1);
    }
};
vector<Vertex> t;
int add_string(string const&s){
    int v = 0;
    for (char ch : s){
        int c = ch - 'a';
        if (t[v].next[c] == -1){
            t[v].next[c] = t.size();
            t.emplace_back(v, ch);
        }
        v = t[v].next[c];
    }
    t[v].leaf = 1;
    return v;
}
int go(int v, char ch);
int get_link(int v){
    if (t[v].link == -1){
        if (v == 0 || t[v].p == 0)
            t[v].link = 0;
        else
            t[v].link = go(get_link(t[v].p), t[v].pch);
    }
    return t[v].link;
}
int get_mlink(int v){
    if (t[v].mlink == -1){
        if (v == 0 || t[v].p == 0)
            t[v].mlink = 0;
        else{
            t[v].mlink = go(get_link(t[v].p), t[v].pch);
            if (t[v].mlink && !t[t[v].mlink].leaf){
                if (t[t[v].mlink].mlink == -1)
                    get_mlink(t[t[v].mlink]);
                t[v].mlink = t[t[v].mlink].mlink;
            }
        }
    }
    return t[v].mlink;
}
int go(int v, char ch){
    int c = ch - 'a';
    if (t[v].go[c] == -1){
        if (t[v].next[c] != -1)
            t[v].go[c] = t[v].next[c];
        else
            t[v].go[c] = v == 0 ? 0 : go(get_link(v), ch);
    }
    return t[v].go[c];
}
```

5.3 Manacher's Algorithm

```
int lps[MAX * 2];
inline void manacher(string &str){
```

```
rep(i, 0, 2*str.size()) lps[i] = 0;
lps[0] = 0;
lps[1]=1;
int l, r, j, k;
rep(i, 2, 2*str.size()){
    l = (i>>1) - (lps[i]>>1);
    r = ((i-1)>>1) + (lps[i]>>1);
    while(1){//widen
        if (l == 0 || r+1 == str.size()) break;
        if (str[l-1] != str[r+1]) break;
        l--, r++;
    }
    lps[i] = r-l+1;
    //jump
    if (lps[i] > 2){
        j = i-1, k = i+1; //while lps[j] inside lps[i]
        while (lps[j] - j < lps[i] - i)
            lps[k++] = lps[j--];
        lps[k] = lps[i] - (i-j); //lps[k] <- edge of lps[i]
        i = k-1; //jump to mirror, which is k
    }
}
return;
```

//Cara dapetin suatu lps(ganjil atau genap)
 //bakal cover dari index-0 berapa sampe berapa
 //maka L = (i-lps[i]>>1), R=(i+lps[i]>>1)-1;
 bool isPalindrome(int l, int r){return lps[r+l+1] >= r-l+1;}

5.4 Z Algorithm

```
vi Z(const string &S){
    vi z(S.size());
    int l = -1, r = -1;
    for (int i = 1; i < S.size(); i++){
        z[i] = i >= r ? 0 : min(r-i, z[i-l]);
        while(i + z[i] < S.size() && S[i+z[i]] == S[z[i]])
            z[i]++;
        if (i + z[i] > r) l = i, r = i+z[i];
    }
    return z;
}
```

5.5 Hashing

```
class hashing{
public:
    int n;
    string s;
    int mod[2] = {1000000007,1000000009};
    vector<ll> key[2];
    vector<pair<ll, ll>> hsh;
    hashing(string _s) : s(_s){
        n = (int) s.size();
        for(int i = 0; i < 2; i++){
            key[i].resize(n+1);
            key[i][0] = 1;
            do{
                key[i][1] = rng() % mod[i];
                if (key[i][1] < 0) key[i][1] += mod[i];
            } while(key[i][1] == 0);
            for (int j = 2; j <= n; j++){
                key[i][j]=1LL * key[i][j-1] * key[i][1] % mod[i];
            }
        }
        hsh.resize(n);
```

```
hsh[0] = {s[0], s[0]};
for (int i = 1; i < n; i++){
    hsh[i].first = (1LL * hsh[i-1].first * key[0][1] <-
        % mod[0] + s[i]) % mod[0];
    hsh[i].second= (1LL * hsh[i-1].second * key[1][1]<-
        % mod[1] + s[i]) % mod[1];
}
}
pair<ll, ll> getHash(int l, int r){
    pair<ll, ll> R = hsh[r];
    if (l == 0){
        return R;
    }
    pair<ll, ll> L = hsh[l-1];
    R.first = (1LL * R.first + mod[0] - 1LL * L.first <-
        key[0][r-l+1] % mod[0]) % mod[0];
    R.second = (1LL * R.second + mod[1] - 1LL * L.second<-
        * key[1][r-l+1] % mod[1]) % mod[1];
    return R;
}
};
```

6 Math

6.1 Extended Euclidean GCD

```
// computes x and y such that ax + by = gcd(a, b) in O(log(min(a, b)))
// returns {gcd(a, b), x, y}
tuple<int, int, int> gcd(int a, int b) {
    if(b == 0) return {a, 1, 0};
    auto [d, x1, y1] = gcd(b, a % b);
    return {d, y1, x1 - y1 * (a / b)};
}
```

6.2 Generalized CRT

```
template<typename T>
T extended_euclid(T a, T b, T& x, T& y) {
    if(b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    T xx, yy, gcd;
    gcd = extended_euclid(b, a % b, xx, yy);
    x = yy;
    y = xx - (yy * (a / b));
    return gcd;
}
template<typename T>
T MOD(T a, T b) {
    return (a % b + b) % b;
}
// return x, lcm. x = a % n && x = b % m
template<typename T>
pair<T, T> CRT(T a, T n, T b, T m) {
    T _n, _m;
    T gcd = extended_euclid(n, m, _n, _m);
    if(n == m) {
        if(a == b)
            return pair<T, T>(a, n);
        else
            return pair<T, T>(-1, -1);
    } else if(abs(a - b) % gcd != 0)
        return pair<T, T>(-1, -1);
    else {
        T lcm = m * n / gcd;
```

```
T x = MOD(a + MOD(n * MOD(_n * ((b - a) / gcd), m / gcd), lcm), lcm);
return pair<T, T>(x, lcm);
}
```

6.3 Generalized Lucas Theorem

```
/*Special Lucas : (n,k) % p^x
fctp[n] = Product of the integers less than or equal
to n that are not divisible by p
Precompute fctp*/
LL p
LL E(LL n, int m) {
    LL tot = 0;
    while(n != 0)
        tot += n / m, n /= m;
    return tot;
}
LL funct(LL n, LL base) {
    LL ans = fast(fctp[base], n / base, base) * fctp[n % base] % base;
    return ans;
}
LL F(LL n, LL base) {
    LL ans = 1;
    while(n != 0) {
        ans = (ans * funct(n, base)) % base;
        n /= p;
    }
    return ans;
}
LL special_lucas(LL n, LL r, LL base) {
    p = fprime(base);
    LL pow = E(n, p) - E(n - r, p) - E(r, p);
    LL TOP = fast(p, pow, base) * F(n, base) % base;
    LL BOT = F(r, base) * F(n - r, base) % base;
    return (TOP * fast(BOT, totien(base) - 1, base)) % base;
}
//End of Special Lucas
```

6.4 Linear Diophantine

```
//FOR SOLVING MINIMUM ABS(X) + ABS(Y)
ll x, y, newX, newY, target = 0;
ll extGcd(ll a, ll b) {
    if(b == 0) {
        x = 1, y = 0;
        return a;
    }
    ll ret = extGcd(b, a % b);
    newX = y;
    newY = x - y * (a / b);
    x = newX;
    y = newY;
    return ret;
}
ll fix(ll sol, ll rt) {
    ll ret = 0;
    //CASE SOLUTION(X/Y) < TARGET
    if(sol < target)
        ret = -floor(abs(sol + target) / (double)rt);
    //CASE SOLUTION(X/Y) > TARGET
    if(sol > target)
        ret = ceil(abs(sol - target) / (double)rt);
    return ret;
}
ll work(ll a, ll b, ll c) {
```

```
ll gcd = extGcd(a, b);
ll solX = x * (c / gcd);
ll solY = y * (c / gcd);
a /= gcd;
b /= gcd;
ll fi = abs(fix(solX, b));
ll se = abs(fix(solY, a));
ll lo = min(fi, se);
ll hi = max(fi, se);
ll ans = abs(solX) + abs(solY);
for(ll i = lo; i <= hi; i++) {
    ans = min(ans, abs(solX + i * b) + abs(solY - i * a));
    ans = min(ans, abs(solX - i * b) + abs(solY + i * a));
}
return ans;
}
```

6.5 Modular Linear Equation

```
// finds all solutions to ax = b (mod n)
vi modular_linear_equation_solver(int a, int b, int n) {
    int x, y;
    vi ret;
    int g = extended_euclid(a, n, x, y);
    if(!(b % g)) {
        x = mod(x * (b / g), n);
        for(int i = 0; i < g; i++)
            ret.push_back(mod(x + i * (n / g), n));
    }
    return ret;
}
```

6.6 Miller-Rabin and Pollard's Rho

```
namespace MillerRabin {
    const vector<ll> primes = { // deterministic up to 2^64 - 1
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37
    };
    ll gcd(ll a, ll b) {
        return b ? gcd(b, a % b) : a;
    }
    ll powa(ll x, ll y, ll p) { // (x ^ y) % p
        if(!y)
            return 1;
        if(y & 1)
            return ((__int128) x * powa(x, y - 1, p)) % p;
        ll temp = powa(x, y >> 1, p);
        return ((__int128) temp * temp) % p;
    }
    bool miller_rabin(ll n, ll a, ll d, int s) {
        ll x = powa(a, d, n);
        if(x == 1 || x == n - 1)
            return 0;
        for(int i = 0; i < s; ++i) {
            x = ((__int128) x * x) % n;
            if(x == n - 1)
                return 0;
        }
        return 1;
    }
    bool is_prime(ll x) { // use this
        if(x < 2)
            return 0;
        int r = 0;
        ll d = x - 1;
```

```
while((d & 1) == 0) {
    d >>= 1;
    ++r;
}
for(auto& i : primes) {
    if(x == i)
        return 1;
    if(miller_rabin(x, i, d, r))
        return 0;
}
return 1;
}
}

namespace PollardRho {
    mt19937_64 generator(chrono::steady_clock::now()
        .time_since_epoch().count());
    uniform_int_distribution<ll> rand_ll(0, LLONG_MAX);
    ll f(ll x, ll b, ll n) { // (x^2 + b) % n
        return ((__int128) x * x) % n + b) % n;
    }
    ll rho(ll n) {
        if(n % 2 == 0)
            return 2;
        ll b = rand_ll(generator);
        ll x = rand_ll(generator);
        ll y = x;
        while(1) {
            x = f(x, b, n);
            y = f(f(y, b, n), b, n);
            ll d = MillerRabin::gcd(abs(x - y), n);
            if(d != 1)
                return d;
        }
    }
    void pollard_rho(ll n, vector<ll>& res) {
        if(n == 1)
            return;
        if(MillerRabin::is_prime(n)) {
            res.push_back(n);
            return;
        }
        ll d = rho(n);
        pollard_rho(d, res);
        pollard_rho(n / d, res);
    }
    vector<ll> factorize(ll n, bool sorted = 1) { // use this
        vector<ll> res;
        pollard_rho(n, res);
        if(sorted)
            sort(res.begin(), res.end());
        return res;
    }
}
```

6.7 Berlekamp-Massey

```
#include <bits/stdc++.h>
using namespace std;
#define pb push_back
typedef long long ll;
#define SZ 233333
const int MOD = 1e9 + 7; //or any prime
ll qp(ll a, ll b) {
    ll x = 1;
    a %= MOD;
    while(b) {
        if(b & 1)
            x = x * a % MOD;
```



```

a = a * a % MOD;
b >>= 1;
}
return x;
}
namespace linear_seq {
vector<int> BM(vector<int> x) {
//ls: (shortest) relation sequence (after filling ←
zeroes) so far
//cur: current relation sequence
vector<int> ls, cur;
//lf: the position of ls (t')
//ld: delta of ls (v')
int lf = -1, ld = -1;
for(int i = 0; i < int(x.size()); ++i) {
ll t = 0;
//evaluate at position i
for(int j = 0; j < int(cur.size()); ++j)
t = (t + x[i - j - 1] * (ll)cur[j]) % MOD;
if((t - x[i]) % MOD == 0) {
continue; //good so far
}
//first non-zero position
if(!cur.size()) {
cur.resize(i + 1);
lf = i;
ld = (t - x[i]) % MOD;
continue;
}
//cur=cur-c/ld*(x[i]-t)
ll k = -(x[i] - t) * qp(ld, MOD - 2) % MOD/*1/ld←
*/;
vector<int> c(i - lf - 1); //add zeroes in front
c.pb(k);
for(int j = 0; j < int(ls.size()); ++j)
c.pb(-ls[j]*k % MOD);
if(c.size() < cur.size())
c.resize(cur.size());
for(int j = 0; j < int(cur.size()); ++j)
c[j] = (c[j] + cur[j]) % MOD;
//if cur is better than ls, change ls to cur
if(i - lf + (int)ls.size() >= (int)cur.size())
ls = cur, lf = i, ld = (t - x[i]) % MOD;
cur = c;
}
for(int i = 0; i < int(cur.size()); ++i)
cur[i] = (cur[i] % MOD + MOD) % MOD;
return cur;
}
int m; //length of recurrence
//a: first terms
//h: relation
ll a[SZ], h[SZ], t_[SZ], s[SZ], t[SZ];
//calculate p*q mod f
void mull(ll* p, ll* q) {
for(int i = 0; i < m + m; ++i)
t_[i] = 0;
for(int i = 0; i < m; ++i) if(p[i])
for(int j = 0; j < m; ++j)
t_[i + j] = (t_[i + j] + p[i] * q[j]) % MOD;
for(int i = m + m - 1; i >= m; --i) if(t_[i])
//miuns t_[i]*x^{i-m}(x^m-∑_{j=0}^{m-1} x^{m-←
j-1}h_j)
for(int j = m - 1; ~j; --j)
t_[i - j - 1] = (t_[i - j - 1] + t_[i] * h[j←
]) % MOD;
for(int i = 0; i < m; ++i)
p[i] = t_[i];
}
ll calc(ll K) {
for(int i = m; ~i; --i)

```

```

s[i] = t[i] = 0;
//init
s[0] = 1;
if(m != 1)
t[1] = 1;
else
t[0] = h[0];
//binary-exponentiation
while(K) {
if(K & 1)
mull(s, t);
mull(t, t);
K >>= 1;
}
ll su = 0;
for(int i = 0; i < m; ++i)
su = (su + s[i] * a[i]) % MOD;
return (su % MOD + MOD) % MOD;
}
int work(vector<int> x, ll n) {
if(n < int(x.size()))
return x[n];
vector<int> v = BM(x);
m = v.size();
if(!m)
return 0;
for(int i = 0; i < m; ++i)
h[i] = v[i], a[i] = x[i];
return calc(n);
}
}
using linear_seq::work;

const vector<int> sequence = {
0, 2, 2, 28, 60, 836, 2766
};
int main() {
cout << work(sequence, 7) << '\n';
}

```

6.8 Fast Fourier Transform

```

using ld = double; // change to long double if reach ←
10^18
using cd = complex<ld>;
const ld PI = acos(-(ld)1);

void fft(vector<cd>& a, int sign = 1) {
int n = a.size();
ld theta = sign * 2 * PI / n;
for(int i = 0, j = 1; j < n - 1; j++) {
for(int k = n >> 1; k > (i ^= k); k >>= 1);
if(j < i)
swap(a[i], a[j]);
}
for(int m, mh = 1; (m = mh << 1) <= n; mh = m) {
int irev = 0;
for(int i = 0; i < n; i += m) {
cd w = exp(cd(0, theta * irev));
for(int k = n >> 2; k > (irev ^= k); k >>= 1);
for(int j = i; j < mh + i; j++) {
int k = j + mh;
cd x = a[j] - a[k];
a[j] += a[k];
a[k] = w * x;
}
}
}
if(sign == -1) for(cd& i : a)
i /= n;

```

```

}
vector<ll> multiply(vector<ll> const& a, vector<ll> ←
const& b) {
vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.←
end());
int n = 1;
while(n < a.size() + b.size())
n <<= 1;
fa.resize(n);
fb.resize(n);
fft(fa);
fft(fb);
for(int i = 0; i < n; i++)
fa[i] *= fb[i];
fft(fa, -1);
vector<ll> res(n);
for(int i = 0; i < n; i++)
res[i] = round(fa[i].real());
return res;
}

```

6.9 Number Theoretic Transform

```

namespace FFT {
/* ----- Adjust the constants here ----- */
const int LN = 24; //23
const int N = 1 << LN;
typedef long long LL; // 2**23 * 119 + 1. 998244353
// 'MOD' must be of the form 2**'LN' * k + 1, where k ←
odd.
const LL MOD = 9223372036737335297; // 2**24 * ←
54975513881 + 1.
const LL PRIMITIVE_ROOT = 3; // Primitive root modulo←
'MOD'.
/* ----- End of constants ----- */
LL root[N];
inline LL power(LL x, LL y) {
LL ret = 1;
for(; y; y >>= 1) {
if(y & 1)
ret = (__int128) ret * x % MOD;
x = (__int128) x * x % MOD;
}
return ret;
}
inline void init_fft() {
const LL UNITY = power(PRIMITIVE_ROOT, MOD - 1 >> ←
LN);
root[0] = 1;
for(int i = 1; i < N; i++)
root[i] = (__int128) UNITY * root[i - 1] % MOD;
return;
}
// n = 2^k is the length of polynom
inline void fft(int n, vector<LL>& a, bool invert) {
for(int i = 1, j = 0; i < n; ++i) {
int bit = n >> 1;
for(; j >= bit; bit >>= 1)
j -= bit;
j += bit;
if(i < j)
swap(a[i], a[j]);
}
for(int len = 2; len <= n; len <<= 1) {
LL wlen = (invert ? root[N - N / len] : root[N / ←
len]);
for(int i = 0; i < n; i += len) {
LL w = 1;
for(int j = 0; j < len >> 1; j++) {

```

```

        LL u = a[i + j];
        LL v = ((__int128) a[i + j + len / 2] * w % MOD;
        MOD;
        a[i + j] = ((__int128) u + v) % MOD;
        a[i + j + len / 2] = ((__int128) u - v + MOD) % MOD;
        w = ((__int128) w * wlen % MOD;
    }
}
}
if(invert) {
    LL inv = power(n, MOD - 2);
    for(int i = 0; i < n; i++)
        a[i] = ((__int128) a[i] * inv % MOD;
}
return;
}
inline vector<LL> multiply(vector<LL> a, vector<LL> b) {
    vector<LL> c;
    int len = 1 << 32 - __builtin_clz(a.size() + b.size() - 2);
    a.resize(len, 0);
    b.resize(len, 0);
    fft(len, a, false);
    fft(len, b, false);
    c.resize(len);
    for(int i = 0; i < len; ++i)
        c[i] = ((__int128) a[i] * b[i] % MOD;
    fft(len, c, true);
    return c;
}
//FFT::init_fft(); wajib di panggil init di awal
}

```

6.10 Gauss-Jordan

```

// Gauss-Jordan elimination with full pivoting.
//
// Uses:
// (1) solving systems of linear equations (AX=B)
// (2) inverting matrices (AX=I)
// (3) computing determinants of square matrices
//
// Running time: O(n^3)
//
// INPUT:  a[][] = an nxn matrix
//         b[][] = an nxm matrix
//
// OUTPUT: X      = an nxm matrix (stored in b[][])
//         A^{-1} = an nxn matrix (stored in a[][])
//         returns determinant of a[][]
const double EPS = 1e-10;

typedef vector<int> VI;
typedef double T;
typedef vector<T> VT;
typedef vector<VT> VVT;
T GaussJordan(VVT& a, VVT& b) {
    const int n = a.size();
    const int m = b[0].size();
    VI irow(n), icol(n), ipiv(n);
    T det = 1;
    for(int i = 0; i < n; i++) {
        int pj = -1, pk = -1;
        for(int j = 0; j < n; j++) if(!ipiv[j])
            for(int k = 0; k < n; k++) if(!ipiv[k])
                if(pj == -1 || fabs(a[j][k]) > fabs(a[pj][pk])) {
                    pk = k;
                    ipiv[pk]++;
                    swap(a[pj], a[pk]);
                    swap(b[pj], b[pk]);
                    if(pj != pk)
                        det *= -1;
                    irow[i] = pj;
                    icol[i] = pk;
                    T c = 1.0 / a[pk][pk];
                    det *= a[pk][pk];
                    a[pk][pk] = 1.0;
                    for(int p = 0; p < n; p++)
                        a[pk][p] *= c;
                    for(int p = 0; p < m; p++)
                        b[pk][p] *= c;
                    for(int p = 0; p < n; p++) if(p != pk) {
                        c = a[p][pk];
                        a[p][pk] = 0;
                        for(int q = 0; q < n; q++)
                            a[p][q] -= a[pk][q] * c;
                        for(int q = 0; q < m; q++)
                            b[p][q] -= b[pk][q] * c;
                    }
                }
        for(int p = n - 1; p >= 0; p--) if(irow[p] != icol[p]) {
            for(int k = 0; k < n; k++)
                swap(a[k][irow[p]], a[k][icol[p]]);
        }
        return det;
    }
}
int main() {
    const int n = 4;
    const int m = 2;
    double A[n][n] = { {1, 2, 3, 4}, {1, 0, 1, 0}, {5, 3, 2, 4}, {6, 1, 4, 6} };
    double B[n][m] = { {1, 2}, {4, 3}, {5, 6}, {8, 7} };
    VVT a(n), b(n);
    for(int i = 0; i < n; i++) {
        a[i] = VT(A[i], A[i] + n);
        b[i] = VT(B[i], B[i] + m);
    }
    double det = GaussJordan(a, b);
    // expected: 60
    cout << "Determinant: " << det << endl;
    // expected: -0.233333 0.166667 0.133333 0.066667
    //              0.166667 0.166667 0.333333 -0.333333
    //              0.233333 0.833333 -0.133333 -0.066667
    //              0.05 -0.75 -0.1 0.2
    cout << "Inverse: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < n; j++)
            cout << a[i][j] << ' ';
        cout << endl;
    }
    // expected: 1.63333 1.3
    //              -0.166667 0.5
    //              2.36667 1.7
    //              -1.85 -1.35
    cout << "Solution: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)
            cout << b[i][j] << ' ';
        cout << endl;
    }
}

```

```

        pk = k;
    }
    if(fabs(a[pj][pk]) < EPS) {
        cerr << "Matrix is singular." << endl;
        exit(0);
    }
    ipiv[pk]++;
    swap(a[pj], a[pk]);
    swap(b[pj], b[pk]);
    if(pj != pk)
        det *= -1;
    irow[i] = pj;
    icol[i] = pk;
    T c = 1.0 / a[pk][pk];
    det *= a[pk][pk];
    a[pk][pk] = 1.0;
    for(int p = 0; p < n; p++)
        a[pk][p] *= c;
    for(int p = 0; p < m; p++)
        b[pk][p] *= c;
    for(int p = 0; p < n; p++) if(p != pk) {
        c = a[p][pk];
        a[p][pk] = 0;
        for(int q = 0; q < n; q++)
            a[p][q] -= a[pk][q] * c;
        for(int q = 0; q < m; q++)
            b[p][q] -= b[pk][q] * c;
    }
}
for(int p = n - 1; p >= 0; p--) if(irow[p] != icol[p]) {
    for(int k = 0; k < n; k++)
        swap(a[k][irow[p]], a[k][icol[p]]);
}
return det;
}
int main() {
    const int n = 4;
    const int m = 2;
    double A[n][n] = { {1, 2, 3, 4}, {1, 0, 1, 0}, {5, 3, 2, 4}, {6, 1, 4, 6} };
    double B[n][m] = { {1, 2}, {4, 3}, {5, 6}, {8, 7} };
    VVT a(n), b(n);
    for(int i = 0; i < n; i++) {
        a[i] = VT(A[i], A[i] + n);
        b[i] = VT(B[i], B[i] + m);
    }
    double det = GaussJordan(a, b);
    // expected: 60
    cout << "Determinant: " << det << endl;
    // expected: -0.233333 0.166667 0.133333 0.066667
    //              0.166667 0.166667 0.333333 -0.333333
    //              0.233333 0.833333 -0.133333 -0.066667
    //              0.05 -0.75 -0.1 0.2
    cout << "Inverse: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < n; j++)
            cout << a[i][j] << ' ';
        cout << endl;
    }
    // expected: 1.63333 1.3
    //              -0.166667 0.5
    //              2.36667 1.7
    //              -1.85 -1.35
    cout << "Solution: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)
            cout << b[i][j] << ' ';
        cout << endl;
    }
}

```

6.11 Josephus Problem

```

ll josephus(ll n, ll k){ //O(klogn)
    if (n == 1) return 0;
    if (k == 1) return n-1;
    if (k > n) return(josephus(n-1,k) + k)%n;
    ll cnt = n/k;
    ll res = josephus(n-cnt, k);
    res -= n%k;
    if (res < 0) res += n;
    else res += res/(k-1);
    return res;
}
int josephus(int n, int k){ //O(n)
    int res = 0;
    for (int i = 1; i <= n; ++i)
        res = (res + k)%i;
    return res+1;
}

```

6.12 Derangement

```

der[0] = 1;
der[1] = 0;
for(int i = 2; i <= 10; ++i)
    der[i] = (1)(i - 1) * (der[i - 1] + der[i - 2]);

```

6.13 Bernoulli Number

$$\sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{i=0}^m \binom{m+1}{i} B_i^+ n^{m+1-i} = m! \sum_{i=0}^m \frac{B_i^+ n^{m+1-i}}{i!(m+1-i)!}$$

$$B_n^+ = 1 - \sum_{i=0}^{n-1} \binom{n}{i} \frac{B_i^+}{n-i+1}, \quad B_0^+ = 1$$

6.14 Forbenius Number

$(X * Y) - (X + Y)$ and total count is $(X - 1) * (Y - 1) / 2$

6.15 Stars and Bars with Upper Bound

$$P = (1 - X^{r_1+1}) \dots (1 - X^{r_n+1}) = \sum_i c_i X^{e_i}$$

$$Ans = \sum_i c_i \binom{N-e_i+n-1}{n-1}$$

6.16 Arithmetic Sequences

$$U_n = a + (n-1)a_1 + \frac{(n-1)(n-2)}{1 \times 2} a_2 + \dots + \frac{(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \times \dots} a_r$$

$$S_n = n \times a + \frac{n(n-1)}{1 \times 2} a_1 + \frac{n(n-1)(n-2)}{1 \times 2 \times 3} a_2 + \dots + \frac{n(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \times \dots} a_r$$

6.17 Mobius Function

For any positive integer n ,

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } a^2 \mid n \text{ for some } a > 1, \\ (-1)^k & \text{if } n \text{ is the product of } k \text{ distinct primes. } \square \end{cases}$$

```
int mobius[N + 1];
void precompute_mobius(){
    for (int i = 1; i <= N; i++) {
        mobius[i] = 1;
    }
    for (int i = 2; i <= N; i++) {
        if (mobius[i] == 1) {
            for (int j = i; j <= N; j += i) {
                if (j % (i * i) == 0) {
                    mobius[j] = 0;
                }
                else {
                    mobius[j] *= -1;
                }
            }
        }
    }
}
```

6.18 Floor Sum

```
// sum_{i=0..n-1} floor((ai+b)/m)
ll sum_of_floor(ll n, ll m, ll a, ll b){
    ll ans = 0;
    if (a >= m){
        ans += (n-1) * n * (a/m) / 2;
        a %= m;
    }
    if (b >= m){
        ans += n * (b/m);
        b %= m;
    }
    ll y_max = (a*n+b)/m, x_max = (y_max*m - b);
    if (y_max == 0) return ans;
    ans += (n - (x_max+a-1)/a) * y_max;
    ans += sum_of_floor(y_max, a, m, (a-x_max%a) % a);
    return ans;
}
```

6.19 Combin Identity

- $\sum_{0 \leq k \leq n} \binom{n-k}{k} = Fib_{n+1}$
- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $k \binom{n}{k} = n \binom{n-1}{k-1}$
- $\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$
- $\sum_{i=0}^n \binom{n}{i} = 2^n$
- $\sum_{i \geq 0} \binom{n}{2i} = 2^{n-1}$
- $\sum_{i \geq 0} \binom{n}{2i+1} = 2^{n-1}$
- $\sum_{i=0}^k (-1)^i \binom{n}{i} = (-1)^k \binom{n-1}{k}$
- $\sum_{i=0}^k \binom{n+i}{i} = \sum_{i=0}^k \binom{n+i}{n} = \binom{n+k+1}{k}$
- $\sum_{i=1}^n i \binom{n}{i} = n 2^{n-1}$
- $\sum_{i=1}^n i^2 \binom{n}{i} = (n + n^2) 2^{n-2}$

$$12. \text{ Vandermonde's Identify: } \sum_{k=0}^m \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

$$13. \text{ Hockey-Stick Identify: } n, r \in \mathbb{N}, n > r, \sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

$$14. \sum_{i=0}^k \binom{k}{i}^2 = \binom{2k}{k}$$

$$15. \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{2n}{n}$$

$$16. \sum_{k=q}^n \binom{n}{k} \binom{k}{q} = 2^{n-q} \binom{n}{q}$$

$$17. \sum_{i=0}^n k^i \binom{n}{i} = (k+1)^n$$

$$18. \sum_{i=0}^n \binom{2n}{i} = 2^{2n-1} + \frac{1}{2} \binom{2n}{n}$$

$$19. \sum_{i=1}^n \binom{n}{i} \binom{n-1}{i-1} = \binom{2n-1}{n-1}$$

$$20. \sum_{i=0}^n \binom{2n}{i}^2 = \frac{1}{2} \left(\binom{4n}{2n} + \binom{2n}{n}^2 \right)$$

21. Number of ways to divide n persons in $\frac{n}{k}$ equal groups i.e. each having size k is

$$\frac{n!}{(k!)^{\frac{n}{k}} \left(\frac{n}{k}\right)!} = \prod_{n \geq k} \binom{n-1}{k-1}^{n-k}$$

$$22. \sum_{i=1,3,5,\dots}^{\leq n} \binom{n}{i} a^{n-i} b^i = \frac{1}{2} ((a+b)^n - (a-b)^n)$$

$$23. \sum_{i=0}^n \frac{\binom{k}{i}}{\binom{n}{i}} = \frac{\binom{n+1}{n-k+1}}{\binom{n}{k}}$$

6.20 Fibonacci Identity

- $F_n = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n-k-1}{k}$
- $F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$
- $\sum_{i=1}^n F_i = F_{n+2} - 1$
- $\sum_{i=0}^{n-1} F_{2i+1} = F_{2n}$
- $\sum_{i=1}^n F_{2i} = F_{2n+1} - 1$
- $\sum_{i=1}^n F_i^2 = F_n F_{n+1}$
- $F_m F_{n+1} - F_{m-1} F_n = (-1)^n F_{m-n}$
- $F_{2n} = F_{n+1}^2 - F_{n-1}^2 = F_n (F_{n+1} + F_{n-1})$
- $F_m F_n + F_{m-1} F_{n-1} = F_{m+n-1}$
- $F_m F_{n+1} + F_{m-1} F_n = F_{m+n}$
- A number is Fibonacci if and only if one or both of $(5 \cdot n^2 + 4)$ or $(5 \cdot n^2 - 4)$ is a perfect square.
- Every third number of the sequence is even and more generally, every k^{th} number of the sequence is a multiple of F_k
- $gcd(F_m, F_n) = F_{gcd(m,n)}$

14. Any three consecutive Fibonacci numbers are pairwise co-prime, which means that, for every n , $gcd(F_n, F_{n+1}) = gcd(F_n, F_{n+2}), gcd(F_{n+1}, F_{n+2}) = 1$

15. The n th Pisano period, written as $\pi(n)$, is the period with which the sequence of Fibonacci numbers taken modulo n repeats.
 $\pi(mn) = lcm(\pi(m), \pi(n))$, with m and n coprime
 If $n = 2^k$, then $\pi(n) = 3 \cdot 2^{k-1} = \frac{3n}{2}$
 If $n = 5^k$, then $\pi(n) = 4 \cdot 5^k = 4n$

7 Graph

7.1 Articulation Point and Bridge

```
//gr -> adjlist
//vector vis, low -> initialize to -1
//int timer -> initialize to 0
void dfs(int pos, int dad = -1){
    vis[pos] = low[pos] = timer++;
    int kids = 0;
    for (auto& i : gr[pos]){
        if (i == dad) continue;
        if (vis[i] >= 0)
            low[pos] = min(low[pos], vis[i]);
        else{
            dfs(i, pos);
            low[pos] = min(low[pos], low[i]);
            if (low[i] > vis[pos])
                is_bridge(pos, i)
            if (low[i] >= vis[pos] && dad >= 0)
                is_articulation_point(pos)
            ++kids;
        }
    }
    if (dad == -1 && kids > 1)
        is_articulation_point(pos)
}
```

7.2 SCC and Strong Orientation

```
#define NMAX
vector<int> v[N];
bool vis[N], ins[N];
int disc[N], low[N], gr[N];
stack<int> st;
int id, grid;
void scc(int cur, int par){
    disc[cur] = low[cur] = ++id;
    vis[cur] = ins[cur] = 1;
    st.push(cur);
    for(int to : v[cur]){
        //if(to==par)continue; //iniuntukSO(sccundirected)
        if (!vis[to])
            scc(to, cur);
        if (ins[to])
            low[cur] = min(low[cur], low[to]);
    }
    if (low[cur] == disc[cur]){
        grid++; //groupid
        while (ins[cur]){
            gr[st.top()] = grid;
            ins[st.top()] = 0;
            st.pop();
        }
    }
}
```

7.3 Centroid Decomposition

```
int build_cen(intn w){
    com_cen(nw, 0); //fungsi untuk itung size subtree
    int siz = sz[nw]/2;
    bool found = false;
    while (!found){
        found = true;
        for (int i : v[nw]){
            if(!rem[i] && sz[i] < sz[nw]){
                if (sz[i] > siz){
                    found = false;
                    nw = i;
                    break;
                }
            }
        }
    }
    rem[nw] = true;
    for (int i : v[nw]) if(!rem[i])
        par_cen[build_cen(i)] = nw;
    return nw;
}
```

7.4 Dinic's Maximum Flow

```
// O(VE log(max_flow)) if scaling == 1
// O((V + E) sqrt(E)) if unit graph (turn scaling off)
// O((V + E) sqrt(V)) if bipartite matching (turn ←
// scaling off)
// indices are 0-based
const ll INF = 1e18;

struct Dinic {
    struct Edge {
        int v;
        ll cap, flow;
        Edge(int _v, ll _cap): v(_v), cap(_cap), flow(0) {}
    };

    int n;
    ll lim;
    vector<vector<int>> gr;
    vector<Edge> e;
    vector<int> idx, lv;

    bool has_path(int s, int t) {
        queue<int> q;
        q.push(s);
        lv.assign(n, -1);
        lv[s] = 0;
        while (!q.empty()) {
            int c = q.front();
            q.pop();
            if(c == t)
                break;
            for(auto& i : gr[c]) {
                ll cur_flow = e[i].cap - e[i].flow;
                if(lv[e[i].v] == -1 && cur_flow >= lim) {
                    lv[e[i].v] = lv[c] + 1;
                    q.push(e[i].v);
                }
            }
        }
        return lv[t] != -1;
    }

    ll get_flow(int s, int t, ll left) {
        if(!left || s == t)
            return left;
        while(idx[s] < (int) gr[s].size()) {
            int i = gr[s][idx[s]];
            if(lv[e[i].v] == lv[s] + 1) {
                ll add = get_flow(e[i].v, t, min(left, e[i].cap ←
                    - e[i].flow));
                if(add) {
                    e[i].flow += add;
                    e[i ^ 1].flow -= add;
                    return add;
                }
            }
            ++idx[s];
        }
        return 0;
    }

    Dinic(int vertices, bool scaling = 1) : // toggle ←
        scaling here
        n(vertices), lim(scaling ? 1 << 30 : 1), gr(n) {}

    void add_edge(int from, int to, ll cap, bool directed ←
        = 1) {
        gr[from].push_back(e.size());
        e.emplace_back(to, cap);
        gr[to].push_back(e.size());
        e.emplace_back(from, directed ? 0 : cap);
    }

    ll get_max_flow(int s, int t) { // call this
        ll res = 0;
        while(lim) { // scaling
            while(has_path(s, t)) {
                idx.assign(n, 0);
                while(ll add = get_flow(s, t, INF))
                    res += add;
            }
            lim >>= 1;
        }
        return res;
    }
};
```

```
while(idx[s] < (int) gr[s].size()) {
    int i = gr[s][idx[s]];
    if(lv[e[i].v] == lv[s] + 1) {
        ll add = get_flow(e[i].v, t, min(left, e[i].cap ←
            - e[i].flow));
        if(add) {
            e[i].flow += add;
            e[i ^ 1].flow -= add;
            return add;
        }
    }
    ++idx[s];
}
return 0;
}

Dinic(int vertices, bool scaling = 1) : // toggle ←
    scaling here
    n(vertices), lim(scaling ? 1 << 30 : 1), gr(n) {}

void add_edge(int from, int to, ll cap, bool directed ←
    = 1) {
    gr[from].push_back(e.size());
    e.emplace_back(to, cap);
    gr[to].push_back(e.size());
    e.emplace_back(from, directed ? 0 : cap);
}

ll get_max_flow(int s, int t) { // call this
    ll res = 0;
    while(lim) { // scaling
        while(has_path(s, t)) {
            idx.assign(n, 0);
            while(ll add = get_flow(s, t, INF))
                res += add;
        }
        lim >>= 1;
    }
    return res;
}
};
```

7.5 Minimum Cost Maximum Flow

```
using FlowT = ll;
using CostT = ll;

const FlowT F_INF = 1e18;
const CostT C_INF = 1e18;
const int MAX_V = 1e5 + 5;
const int MAX_E = 1e6 + 5;

namespace MCMF {
    int n, E;
    int adj[MAX_E], nxt[MAX_E], lst[MAX_V], frn[MAX_V], ←
        vis[MAX_V];
    FlowT cap[MAX_E], flw[MAX_E], totalFlow;
    CostT cst[MAX_E], dst[MAX_V], totalCost;

    void init(int _n) {
        n = _n;
        fill_n(lst, n, -1), E = 0;
    }

    void add(int u, int v, FlowT ca, CostT co) {
        adj[E] = v, cap[E] = ca, flw[E] = 0, cst[E] = +co;
        nxt[E] = lst[u], lst[u] = E++;
        adj[E] = u, cap[E] = 0, flw[E] = 0, cst[E] = -co;
        nxt[E] = lst[v], lst[v] = E++;
    }

    int spfa(int s, int t) {
```

```
fill_n(dst, n, C_INF), dst[s] = 0;
queue<int> que;
que.push(s);
while(que.size()) {
    int u = que.front();
    que.pop();
    for(int e = lst[u]; e != -1; e = nxt[e])
        if(flw[e] < cap[e]) {
            int v = adj[e];
            if(dst[v] > dst[u] + cst[e]) {
                dst[v] = dst[u] + cst[e];
                frn[v] = e;
                if(!vis[v]) {
                    vis[v] = 1;
                    que.push(v);
                }
            }
        }
    vis[u] = 0;
}
return dst[t] < C_INF;
}

pair<FlowT, CostT> solve(int s, int t) {
    totalCost = 0, totalFlow = 0;
    while(1) {
        if(!spfa(s, t))
            break;
        FlowT mn = F_INF;
        for(int v = t, e = frn[v]; v != s; v = adj[e ^ ←
            1], e = frn[v])
            mn = min(mn, cap[e] - flw[e]);
        for(int v = t, e = frn[v]; v != s; v = adj[e ^ ←
            1], e = frn[v]) {
            flw[e] += mn;
            flw[e ^ 1] -= mn;
        }
        totalFlow += mn;
        totalCost += mn * dst[t];
    }
    return {totalFlow, totalCost};
}
};
```

7.6 Flows with Demands

let S_0 be the source and T_0 be the original sink

- add 2 additional nodes, call them S_1 and T_1
- connect S_0 to nodes normally
- connect nodes to T_0 normally
- for each edge (U, V) , $\text{cap} = \text{original cap} - \text{demand}$
- for each node N :
 - add an edge (S_1, N) , $\text{cap} = \text{sum of inward demand to } N$
 - add an edge (N, T_1) , $\text{cap} = \text{sum of outward demand } \leftarrow$ from N
- add an edge (T_0, S_0) , $\text{cap} = \text{INF}$
- the above is not a typo!
- run max flow normally
- for each edge (S_1, V) and (U, T_1) , check if $\text{flow} == \leftarrow$ cap

if step #9 fails, then it is not possible to satisfy $\leftarrow$ the given demand

Mathematically, let $d(e)$ be the demand of edge e . Let V be the set of every vertex in the graph.

- $c'(S_1, v) = \sum_{u \in V} d(u, v)$ for each edge (s', v) .
- $c'(v, T_1) = \sum_{v \in V} d(v, w)$ for each edge (v, t') .

- $c'(u, v) = c(u, v) - d(u, v)$ for each edge (u, v) in the old network.
- $c'(T_0, S_0) = \infty$

7.7 Hungarian

```
template <typename TD> struct Hungarian {
    TD INF = 1e9; //max_inf
    int n;
    vector<vector<TD>> > adj; // cost[left][right]
    vector<TD> hl, hr, slk;
    vector<int> fl, fr, vl, vr, pre;
    deque<int> q;
    Hungarian(int _n) {
        n = _n;
        adj = vector<vector<TD>> >(n, vector<TD> (n, 0));
    }
    int check(int i) {
        if(vl[i] == 1, fl[i] != -1)
            return q.push_back(fl[i]), vr[fl[i]] = 1;
        while(i != -1)
            swap(i, fr[fl[i] = pre[i]]);
        return 0;
    }
    void bfs(int s) {
        slk.assign(n, INF);
        vl.assign(n, 0);
        vr = vl;
        q.assign(vr[s] = 1, s);
        for(TD d;;) {
            for(; !q.empty(); q.pop_front()) {
                for(int i = 0, j = q.front(); i < n; i++) {
                    if(d = hl[i] + hr[j] - adj[i][j], !vl[i] && d <= slk[i]) {
                        if(pre[i] = j, d)
                            slk[i] = d;
                        else if(!check(i))
                            return;
                    }
                }
            }
            d = INF;
            for(int i = 0; i < n; i++) if(!vl[i] && d > slk[i])
                d = slk[i];
            for(int i = 0; i < n; i++) {
                if(vl[i])
                    hl[i] += d;
                else
                    slk[i] -= d;
                if(vr[i])
                    hr[i] -= d;
            }
            for(int i = 0; i < n; i++) if(!vl[i] && !slk[i] <-&
                && !check(i))
                return;
        }
    }
    TD solve() {
        fl.assign(n, -1);
        fr = fl;
        hl.assign(n, 0);
        hr = hl;
        pre.assign(n, 0);
        for(int i = 0; i < n; i++)
            hl[i] = *max_element(adj[i].begin(), adj[i].begin()
                () + n);
        for(int i = 0; i < n; i++)
            bfs(i);
    }
};
```

```
TD ret = 0;
for(int i = 0; i < n; i++) if(adj[i][fl[i]])
    ret += adj[i][fl[i]];
return ret;
}; //i will be matched with fl[i]
```

7.8 Edmond's Blossom

```
// Maximum matching on general graphs in O(V^2 E)
// Indices are 1-based
// Stolen from ko_osaga's cheatsheet
struct Blossom {
    vector<int> vis, dad, orig, match, aux;
    vector<vector<int>> conn;
    int t, N;
    queue<int> Q;

    void augment(int u, int v) {
        int pv = v;
        do {
            pv = dad[v];
            int nv = match[pv];
            match[v] = pv;
            match[pv] = v;
            v = nv;
        } while(u != pv);
    }

    int lca(int v, int w) {
        ++t;
        while(true) {
            if(v) {
                if(aux[v] == t)
                    return v;
                aux[v] = t;
                v = orig[dad[match[v]]];
            }
            swap(v, w);
        }
    }

    void blossom(int v, int w, int a) {
        while(orig[v] != a) {
            dad[v] = w;
            w = match[v];
            if(vis[w] == 1) {
                Q.push(w);
                vis[w] = 0;
            }
            orig[v] = orig[w] = a;
            v = dad[w];
        }
    }

    bool bfs(int u) {
        fill(vis.begin(), vis.end(), -1);
        iota(orig.begin(), orig.end(), 0);
        Q = queue<int>();
        Q.push(u);
        vis[u] = 0;
        while(!Q.empty()) {
            int v = Q.front();
            Q.pop();
            for(int x : conn[v]) {
                if(vis[x] == -1) {
                    dad[x] = v;
                    vis[x] = 1;
                    if(!match[x]) {
                        augment(u, x);
                    }
                }
            }
        }
    }
};
```

```
        return 1;
    }
    Q.push(match[x]);
    vis[match[x]] = 0;
} else if(vis[x] == 0 && orig[v] != orig[x]) {
    int a = lca(orig[v], orig[x]);
    blossom(x, v, a);
    blossom(v, x, a);
}
}
}
return false;
}

Blossom(int n) : // n = vertices
    vis(n + 1), dad(n + 1), orig(n + 1), match(n + 1),
    aux(n + 1), conn(n + 1), t(0), N(n) {
    for(int i = 0; i <= n; ++i) {
        conn[i].clear();
        match[i] = aux[i] = dad[i] = 0;
    }
}

void add_edge(int u, int v) {
    conn[u].push_back(v);
    conn[v].push_back(u);
}

int solve() { // call this for answer
    int ans = 0;
    vector<int> V(N - 1);
    iota(V.begin(), V.end(), 1);
    shuffle(V.begin(), V.end(), mt19937(0x94949));
    for(auto x : V) {
        if(!match[x]) {
            for(auto y : conn[x]) {
                if(!match[y]) {
                    match[x] = y, match[y] = x;
                    ++ans;
                    break;
                }
            }
        }
        for(int i = 1; i <= N; ++i) {
            if(!match[i] && bfs(i))
                ++ans;
        }
        return ans;
    }
};
```

7.9 Eulerian Path or Cycle

```
// finds a eulerian path / cycle
// visits each edge only once
// properties:
// - cycle: degrees are even
// - path: degrees are even OR degrees are even except <-&
//         for 2 vertices
// how to use: g = adjacency list g[n] = connected to n<-&
//             , undirected
// if there is a vertex u with an odd degree, call dfs(<-&
// u)
// else call on any vertex
// ans = path result

vector<set<int>> g;
vector<int> ans;
```



```
void dfs(int u) {
    while(g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        g[v].erase(u);
        dfs(v);
    }
    ans.push_back(u);
}
//Make sure that ans.size() == edges + 1 !!!
```

7.10 Hierholzer's Algorithm

```
// Eulerian on Directed Graph
stack<int> path;
vector<int> euler;
inline void hierholzer() {
    path.push(0);
    int cur = 0;
    while(!path.empty()) {
        if(!adj[cur].empty()) {
            path.push(cur);
            int next = adj[cur].back();
            adj[cur].pop();
            cur = next;
        } else {
            euler.pb(cur);
            cur = path.top();
            path.pop();
        }
    }
    reverse(euler.begin(), euler.end());
}
```

7.11 2-SAT

```
struct TwoSAT {
    int n;
    vector<vector<int>> g, gr;
    vector<int> comp, topological_order, answer;
    vector<bool> vis;

    TwoSAT() {}
    TwoSAT(int _n) :
        n(_n), g(2 * n), gr(2 * n), comp(2 * n), answer(2 * n), vis(2 * n) {}

    void add_edge(int u, int v) {
        g[u].push_back(v);
        gr[v].push_back(u);
    }

    // For the following three functions
    // int x, bool val: if 'val' is true, we take the ←
    // variable to be x.
    // Otherwise we take it to be x's complement.

    // At least one of them is true
    void add_clause_or(int i, bool f, int j, bool p) {
        add_edge(i + (f ? n : 0), j + (p ? 0 : n));
        add_edge(j + (p ? n : 0), i + (f ? 0 : n));
    }

    // Only one of them is true
    void add_clause_xor(int i, bool f, int j, bool p) {
        add_clause_or(i, f, j, p);
        add_clause_or(i, !f, j, !p);
    }
}
```

```
// Both of them have the same value
void add_clause_and(int i, bool f, int j, bool p) {
    add_clause_xor(i, !f, j, p);
}

// Topological sort
void dfs(int u) {
    vis[u] = true;
    for(const auto& v : g[u])
        if(!vis[v])
            dfs(v);
    topological_order.push_back(u);
}

// Extracting strongly connected components
void scc(int u, int id) {
    vis[u] = true;
    comp[u] = id;
    for(const auto& v : gr[u])
        if(!vis[v])
            scc(v, id);
}

bool satisfiable() {
    fill(vis.begin(), vis.end(), false);
    for(int i = 0; i < 2 * n; i++)
        if(!vis[i])
            dfs(i);
    fill(vis.begin(), vis.end(), false);
    reverse(topological_order.begin(), ←
            topological_order.end());
    int id = 0;
    for(const auto& v : topological_order)
        if(!vis[v])
            scc(v, id++);
    // Constructing the answer
    for(int i = 0; i < n; i++) {
        if(comp[i] == comp[i + n])
            return false;
        answer[i] = (comp[i] > comp[i + n] ? 1 : 0);
    }
    return true;
}
};
```

7.12 Edge Coloring

```
void colorEdges(int ptr, vector<vector<pair<int, int>>>& gra, vector<int>& edgeColors, bool isVisited←
[])
{
    queue<int> q;
    int c = 0;
    unordered_set<int> colored;
    if (isVisited[ptr]) return;

    isVisited[ptr] = 1;
    for (int i = 0; i < gra[ptr].size(); i++) {
        if (edgeColors[gra[ptr][i].second] != -1)
            colored.insert(edgeColors[gra[ptr][i].second]);
    }

    for (int i = 0; i < gra[ptr].size(); i++) {
        if (!isVisited[gra[ptr][i].first])
            q.push(gra[ptr][i].first);
        if (edgeColors[gra[ptr][i].second] == -1) {
            while (colored.find(c) != colored.end())
                c++;
            edgeColors[gra[ptr][i].second] = c;
            colored.insert(c);
        }
    }
}
```

```
c++;
    }
}
while (!q.empty()) {
    int temp = q.front();
    q.pop();
    colorEdges(temp, gra, edgeColors, isVisited);
}
}
```

7.13 K-th Shortest Path

```
// k = shortest path ke berapa.
// n = jumlah vertex.
// solve(i,j): shortest path dari i ke j.
const int MAXN = 100;
const int MAXK = 40;
vector<pair<int,int>> g[MAXN];
priority_queue<pair<int,int>> pq;
int visit[MAXN], n, k;
int solve(int s, int t){
    FILL(visit, 0);
    pq.push(mp(0, s));
    int ret = 0;
    while (!pq.empty()){
        int awal = -pq.top().first, f = pq.top().second;
        pq.pop();
        if (visit[f] >= k) continue;
        visit[f]++;
        if (f == t) ret += awal;
        for (auto x : g[f]) if (visit[x.first] < k)
            pq.push(mp(-(awal + x.se), x.first));
    }
    return ret;
}
int inside_main(){
    k = GI;
    n = GI;
    for (int nedge = 0; nedge < GI; nedge++){
        int a = GI-1, b = GI-1, c = GI;
        g[a].pb(mp(b,c));
        g[b].pb(mp(a,c));
    }
    printf("%d\n",solve(0,n-1));
}
```

8 OEIS

8.1 A000108 (Catalan Number)

```
//C(2n,n)/(n+1) == C(2n,n)-C(2n,n-1)
//1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, ←
58786, 208012
int findCatalan(int n){
    int catalan[n + 1];
    catalan[0] = catalan[1] = 1;
    for (int i = 2; i <= n; i++){
        catalan[i] = 0;
        for (int j = 0; j < i; j++)
            catalan[i] += catalan[j] * catalan[i - j - 1];
    }
    return catalan[n];
}

//C(2m,m)*C(2n,n)/C(m+n,m)
//1,1,3,11,45,197,903,4279,20793,103049,518859,2646723
ll superCat(int n){
    if (n <= 1) return 1;
```

```
return (3 * (2*n - 1) * superCat(n-1) -
        (n-2) * superCat(n-2)) / (n+1);
}
```

8.2 A000110 (Bell Numbers)

Total number of partitions of n distinct elements.
 $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$
For p prime,
 $B(p^m + n) \equiv mB(n) + B(n + 1) \pmod{p}$

8.3 A000127

Maximal number of regions obtained by joining n points around a circle by straight lines.
 $f(n) = (n^4 - 6n^3 + 23n^2 - 18n + 24)/24$
1, 2, 4, 8, 16, 31, 57, 99, 163, 256, 386, 562, 794, 1093, 1471, 1941, 2517, 3214, 4048, 5036, 6196, 7547, 9109, 10903, 12951, 15276, 17902, 20854, 24158, 27841, 31931, 36457, 41449, 46938, 52956, 59536, 66712, 74519, 82993, 92171, 102091, 112792, 124314

8.4 A001434

Number of graphs with n nodes and n edges
0, 0, 1, 2, 6, 21, 65, 221, 771, 2769, 10250, 39243, 154658, 628635, 2632420, 11353457, 50411413, 230341716, 1082481189, 5228952960, 25945377057, 132140242356, 690238318754

8.5 A018819

Binary partition function: number of partitions of n into powers of 2
 $f(2m + 1) = f(2m); f(2m) = f(2m - 1) + f(m)$
1, 1, 2, 2, 4, 4, 6, 6, 10, 10, 14, 14, 20, 20, 26, 26, 36, 36, 46, 46, 60, 60, 74, 74, 94, 94, 114, 114, 140, 140, 166, 166, 202, 202, 238, 238, 284, 284, 330, 330, 390, 390, 450, 450, 524, 524, 598, 598, 692, 692, 786, 786, 900, 900, 1014, 1014, 1154, 1154, 1294, 1294

8.6 A092098

```
//3-Portolan numbers: number of regions formed by
//n-secting the angles of an equilatera ltriangle.
ll solve(ll n){
    ll res = (n%2==1 ? 3*n*n - 3*n+1 : 3*n*n - 6*n + 6)←
        ;
    const int bats = n/2 - 1;
    for(ll i = 1; i <= bats; i++){
        for(ll j = 1; j <= bats; j++){
            ll num = i*(n-j)*n;
            ll denum = (n-i)*j + i*(n-j);
            res -= 6*(num % denum == 0 && num/denum <= ←
                bats);
        }
    }
    return res;
}
//1,6,19,30,61,78,127,150,217,246,331,366,469,510,
//625,678,817,870,1027,1080,1261,1326,1519,1566,1801,
//1878,2107,2190,2437,2520,2791,2886,3169,3270,3559,
//3678,3997,4110,4447,4548,4921,5034,5419,5550,5899,
//6078,6487
```